Master of Cloud Computing and Internet of Things

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MOODUP – A USER STUDY OF AN EMOTION-AWARE AMBIENT LIGHTING SYSTEM USING SENTIMENT ANALYSIS OF DIGITAL JOURNAL ENTRIES

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REZUMAT

Stresul si situațiile solicitante din punct de vedere emoțional au devenit din ce în ce mai frecvente în viața de zi cu zi. Acest lucru motivează cercetarea unor tehnologii care susțin reflectarea asupra emoțiilor. De asemenea, factori precum temperatura, umiditatea, intensitatea luminii și culoarea ei, pot influența emoțiile și starea de bine a oamenilor. Această disertație prezintă “MoodUp”, un sistem de iluminat ambiental receptiv la emoții, care adaptează culorile luminii în funcție de starea emoțională a utilizatorului, detectată din textul scris într-un jurnalului digital. Pe lângă implementarea sistemului, această lucrare analizează și modul în care utilizatorii percep un mediu de iluminat adaptiv care răspunde emoțiilor, în comparație cu un mediu cu lumină statică tradițională.

MoodUp este alcătuit dintr-o aplicație mobilă dezvoltată în Kotlin și un dispozitiv inteligent bazat pe ESP32. Sistemul poate monitoriza condițiile de mediu și poate adapta dinamic culorile iluminatului ambiental în funcție de tonul emoțional detectat în jurnalul utilizatorului, folosind un algoritm de analiză a sentimentelor procesat local. În cadrul unui studiu cu măsurători repetate, la care au participat 17 persoane, s-a comparat condiția de iluminat existentă a camerei cu sistemul propus de iluminat adaptiv și receptiv la emoții. Percepțiile participanților au fost cuantificate prin intermediul unui chestionar format din zece întrebări, urmat de evaluarea prin “System Usability Scale” (SUS) și de feedback opțional.

Scenariul cu iluminare adaptivă a înregistrat rezultate mai mari pentru toate întrebările chestionarului, în special pentru aspectele legate de receptivitatea mediul, adaptarea în funcție de sentimente și sprijinul perceput în timpul activității de notare în jurnal. MoodUp a obținut un scor SUS de 90,74/100, ceea ce indică un nivel excelent de uzabilitate percepută. De asemenea, 82,4% dintre participanți au considerat combinația dintre aplicația mobilă și dispozitivul de iluminat adaptiv drept cea mai valoroasă parte a sistemului.

În cele din urmă, rezultatele arată faptul că integrarea activității de scriere într-un jurnal digital cu iluminatul ambiental adaptiv creează o experiență mai receptivă și mai favorabilă din punct de vedere emoțional decât iluminatul static tradițional. Sistemul propus demonstrează că adaptarea mediului pe baza emoțiilor poate fi integrată cu succes într-o singură platformă IoT și oferă dovezi că utilizatorii percep pozitiv acest tip de mediu adaptiv.

ABSTRACT

Stress and emotionally demanding situations have become increasingly common in day-to-day life. This motivates research into technologies that support emotional awareness and self-reflection. Also, environmental factors such as temperature, humidity, light intensity and colour can influence human emotions and wellbeing. This dissertation presents MoodUp, an emotion-aware ambient lighting system that adjusts lighting colours according to the user's emotional state detected from digital journal entries. Besides implementing the system, this work also looks into how users perceive an adaptive lighting environment that responds to emotions in comparison with a traditional static light.

MoodUp consists of a Kotlin-based mobile application and an ESP32-based smart lighting device. It can monitor environmental conditions and dynamically adapt ambient light colours according to the emotional tone detected from user-written journal entries, using a locally processed sentiment analysis algorithm. A within-subject user study involving 17 participants compared an existing room lighting condition with the proposed adaptive emotion-aware lighting condition. Participant perceptions were quantified using a ten-item questionnaire, followed by a System Usability Scale evaluation and optional feedback.

The adaptive lighting condition received higher ratings across all questionnaire items, particularly for environmental responsiveness, emotional adaptation and perceived support during journaling. MoodUp achieved a SUS score of 90.74/100, which indicates excellent perceived usability. 82.4% of participants considered the combination of the mobile application and adaptive lighting device to be the most valuable aspect of the system.

Overall, the results indicate that integrating digital journaling with adaptive ambient lighting creates a more responsive and emotionally supportive experience than traditional static lighting. The proposed system demonstrates that emotion-aware environmental adaptation can be successfully integrated within a single IoT platform and provides initial evidence that users perceive this type of adaptive environment positively.

CONTENTS

LIST OF ABBREVIATIONS

AI	Artificial Intelligence
API	Application Programming Interface
DHT	Digital Humidity and Temperature
HCI	Human-Computer Interaction
IoT	Internet of Things
LED	Light-Emitting Diode
MVVM	Model-View-ViewModel
NLP	Natural Language Processing
RGB	Red, Green, Blue
SUS	System Usability Scale
UI	User Interface

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1. INTRODUCTION

1.1 CONTEXT

In the current digital era, which promises accessibility and comfort, people spend more time indoors than ever before. Whether at home, at school, at work, in restaurants, while shopping, etc., individuals prefer an enclosed environment rather than spending time outdoors. Figure 1.1 shows the percentage of annual time spent indoors versus outdoors over the past few years [1].

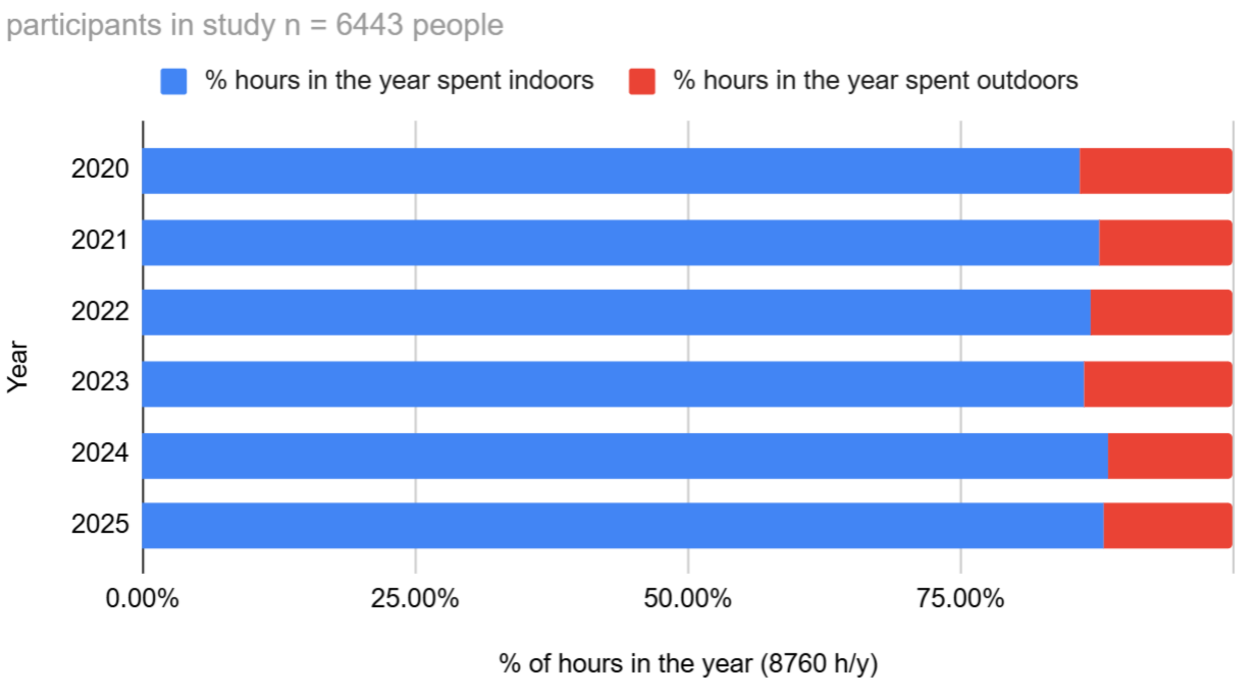

Figure 1.1 - Percentages of Annual Time Spent Indoors vs Outdoors by Year [1]

This being said, people should pay more attention to the environments in which they spend the majority of their time. The quality of indoor spaces has a direct impact on their occupants, influencing comfort, mood, productivity, and overall wellbeing. Factors like temperature, humidity, lighting intensity or colour, noise levels, room proportions, etc. influence how people perceive and interact with their surroundings [2].

One environmental factor that deserves particular attention is lighting. From brightness and intensity to colour and shade, lighting directly influences how people experience and interact with indoor spaces. Poor lighting conditions can affect circadian rhythms and increase fatigue levels, but appropriate lighting can create good sleeping patterns and improve comfort [3]. The advances related to smart lighting technologies have made it possible to adjust lighting based on user preferences. They show a promising way of creating more personalised indoor environments based on specific individual needs with the end goal of improving human wellbeing.

One other aspect that can help improve emotional wellbeing is the ability of individuals to reflect on their own thoughts and feelings. Stress and anxiety are too common in this fast-paced modern society. In order to better understand and process their emotions, many people resort to reflective practices such as journaling. Writing emotions and feelings down can help individuals organise their own thoughts and become more aware of their own wellbeing. This practice has been done on pen and paper over centuries, but modern digital journaling applications have improved the practice and support emotional tracking and mood analysis [4].

Recent developments in the Internet of Things have created new opportunities for combining user-centred applications with intelligent environments. With the integration of sensors and adaptive control systems, it is now possible to create spaces that respond to the needs of users in real time. These tools continue to improve people's day-to-day lives as technology continues to evolve rapidly and inevitably.

This dissertation explores an adaptive ambient lighting system called MoodUp that uses a sentiment analysis algorithm on journal entries written by users and adjusts lighting conditions based on the detected emotional tone. The proposed system also offers mood visualisation features and personalised feedback that encourages emotional self-reflection. It can also monitor the environmental factors (temperature, humidity, lighting), providing additional insights to the user. The end goal of MoodUp is to create an adaptive environment designed to encourage emotional reflection and provide wellbeing support. After the system's implementation, a user study was conducted to evaluate its effectiveness by comparing a traditional static lighting environment with an emotion-aware one, to which participants responded positively.

1.2 PROBLEM STATEMENT

Smart home technologies have evolved significantly over the past few years. Modern lighting systems are capable of many things, like changing brightness, colour or intensity based on user preferences or automation rules. But despite these advantages, the majority of lighting systems still require direct input from the user and do not take into account the emotional state of the user whatsoever.

At the same time, while pen and paper are still a viable practice, digital wellbeing applications are slowly taking over the practice of journaling. They offer many features, from writing about daily events to complex mood visualisations. While they can help users become more aware of their feelings, the collected information rarely influences the surrounding environment in which the user spends most of their time.

Keeping these two concepts in mind, there is a disconnect between physical spaces and emotional self-reflection. Smart environments can automatically respond to external factors and act accordingly, but they rarely adapt based on the emotional state of the user. Similarly, users may record emotions and track their wellbeing digitally, but the environment around them does not change unless they perform the actions themselves.

This creates the opportunity to investigate if a more responsive indoor environment can be created based on emotional information obtained through journaling. And of course, this raises the question of how users would perceive an adaptation like this and if they would consider it useful and practical.

To address this gap, this dissertation presents MoodUp, which is an adaptive ambient lighting system that changes according to the user's emotions detected through a sentiment analysis algorithm based on their journaling entries. A user study was also conducted to demonstrate the practicality and effectiveness of this system, leading to results that reflect the overall user experience.

1.3 AREAS OF STUDY

This dissertation combines concepts from several research areas described below.

IoT provides the technological foundation of the proposed system through the integration of sensors, connected devices and the capability to monitor indoor environments. IoT technologies enable the collection of contextual information and facilitate adaptive control of indoor environments.

Ambient Intelligence and Smart Environments focus on the development of context-aware systems that can respond to user needs and changes in the surrounding environment. These technologies enable the creation of adaptive spaces designed to improve user comfort and overall experience.

Environmental Psychology and Wellbeing research provide insights into to the relationship between environmental factors and human emotions. Previous studies investigating the relationship between lighting, mood and perceived comfort form an important foundation for this work.

Sentiment Analysis is utilised to extract emotional information from journal entries written by users. By analysing textual content, the system can identify emotional tendencies and generate adaptive responses through environmental modifications.

Human-Computer Interaction informs the evaluation methodology adopted in this dissertation. User-centred evaluation techniques are employed to investigate participant perceptions, usability and overall experience with the proposed adaptive environment.

Finally, Affective Computing contributes principles related to systems capable of recognising and responding to human emotions. In MoodUp, emotional information extracted from journal entries is used to generate adaptive environmental responses.

1.4 RESEARCH QUESTIONS

This dissertation is guided by the following research questions:

RQ1: How can a sentiment-aware IoT-based ambient lighting system be designed and implemented using digital journal entries as a source of emotional information?

RQ2: How do users perceive an adaptive ambient lighting environment that responds to emotions expressed through digital journaling?

RQ3: Is an adaptive ambient lighting environment perceived more positively than a traditional static lighting environment?

RQ4: To what extent do users perceive MoodUp as a useful and usable tool for emotional self-reflection and wellbeing support?

1.5 MOTIVATION AND OBJECTIVES

The motivation behind this research comes from the growing presence of stress and emotionally demanding situations in day-to-day life. The academic pressure, the work environment, or personal situations can affect people's emotional wellbeing considerably. Although professional psychological support remains an important part of mental healthcare, many individuals do not actually look for it because of factors like accessibility, cost, stigma or simply not knowing when or where to ask for help.

At the same time, advances in technology have made it now possible to create systems that can respond to user needs in real time. Advances in Internet of Things technologies have made it possible to build adaptive environments capable of monitoring contextual information and reacting to it. Keeping that in mind, it becomes interesting to explore whether environmental adaptation can contribute to how people experience a space and interact with it.

This dissertation does not aim to replace professional therapy or provide medical advice. Instead, it investigates whether adaptive environmental feedback can act as a supportive element that encourages emotional reflection and awareness. Particular attention is given to digital journaling as a way for people to express emotions, look back on their experiences and recognise patterns that may emerge over time.

The main objective of this dissertation is to design and evaluate MoodUp, an adaptive ambient lighting system that uses sentiment analysis of digital journal entries to personalise indoor lighting conditions. Additional objectives include integrating environmental monitoring features, visualising emotional information over time and evaluating user perceptions through an experimental study. By comparing an adaptive lighting condition with a traditional static lighting condition, this research investigates how users experience and perceive an emotion-aware ambient environment.

1.6 PERSONAL CONTRIBUTIONS

The personal contribution of this dissertation includes both the development of MoodUp, and the design of the evaluation methodology used to investigate user perceptions of the proposed system. An overview of the author's main contributions is presented in Figure 1.2.

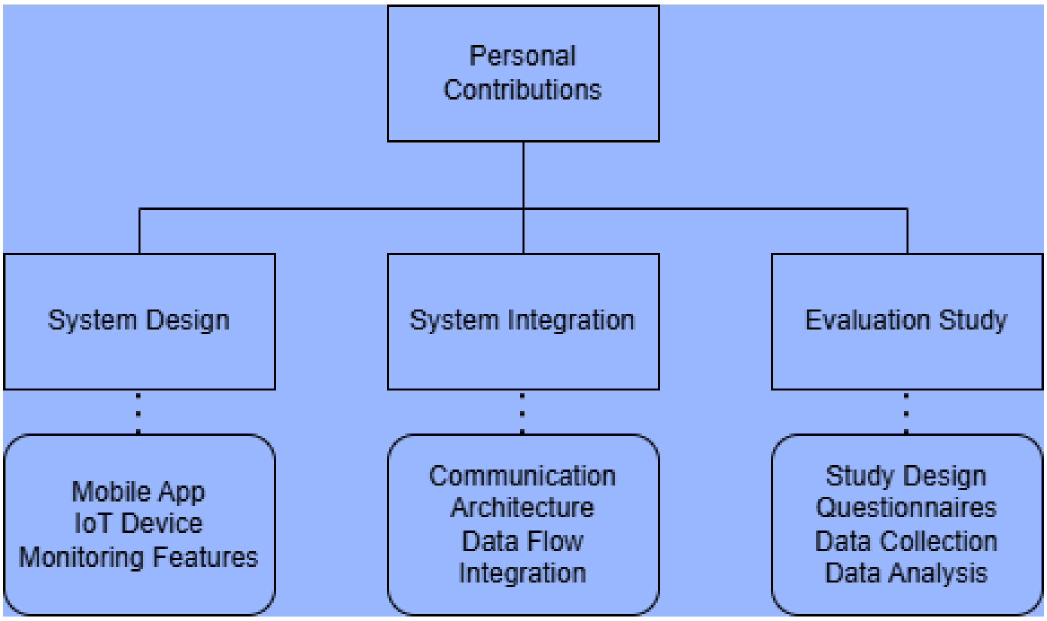

Figure 1.2 – Overview of the Author's Main Contributions

From a technical perspective, the author designed and implemented an IoT-based adaptive ambient lighting system consisting of a mobile application and a smart lighting device capable of monitoring environmental parameters. The developed solution combines digital journaling, sentiment analysis and adaptive lighting responses within a single system.

In addition to the implementation of the system itself, the author designed the communication architecture between the hardware and software components, configured the environmental monitoring infrastructure and integrated user interaction features into the mobile application and smart lighting system.

Another contribution of this work is the design and execution of a user-centred evaluation study. This included developing the questionnaires, defining the experimental procedure, collecting participant feedback and analysing the results. The study provides insight into how users perceive adaptive ambient lighting environments and whether sentiment-aware systems can support emotional reflection and awareness.

1.7 RELATIONSHIP TO PREVIOUS BACHELOR THESIS

This dissertation is a continuation and extension of the author's Bachelor Thesis entitled "MOODUP - A Smart Ambient Lighting System Using Sentiment Analysis Controlled by a Mobile Chatting Application" [5]. The main differences between the original project and the current dissertation are summed up in Table 1.1.

Table 1.1 - Differences Between the Bachelor Thesis and Current Dissertation MoodUp

Bachelor Thesis	Current Dissertation
Conversational AI Assistant	Digital journaling
Cloud-based chat messages sentiment analysis	Local journal entry sentiment analysis
Motion-activated adaptive ambient lighting	Tap-activated adaptive ambient lighting
Environmental Monitoring	User suggestions based on environmental monitoring
	Refined hardware architecture
	Structured user evaluation study
	Comparative testing scenarios and System Usability Scale evaluation
	Participant Feedback Analysis

The original project focused primarily on the design and implementation of a smart ambient lighting system that can adjust lighting conditions according to emotions detected from user interactions. The system incorporated a conversational AI component through which users communicated with a virtual assistant. Sentiment analysis was performed on the sent messages, and the resulting emotional classification was used to modify the lighting environment. The system also included environmental monitoring through a set of sensors and cloud-based data management services.

The present dissertation keeps the fundamental idea of sentiment-aware environmental adaptation while introducing a few changes. Rather than relying on conversations with an AI assistant, the new system brings a journaling-based approach that encourages users to record their thoughts and emotions directly. This design emphasises on emotional reflection and to reduce the influence of AI-generated responses on participant perceptions during the evaluation.

The sentiment analysis component was simplified and modified to be closer to the objectives of the study. This approach improves portability, reduces dependency on network connectivity and external APIs, and increases system reliability during the evaluation sessions. The mobile application was modified to include mood history visualisation, environmental recommendations and small cheering messages. Also, the hardware platform was refined and rebuilt using newer components. Environmental monitoring was expanded and an additional user interaction feature was added to the device.

The most significant extension beyond the Bachelor Thesis is the introduction of a structured user evaluation study. While the previous work focused on the technicalities of designing and developing MoodUp, the present dissertation investigates how users perceive and interact with an adaptive ambient lighting environment through comparative evaluation scenarios, usability assessment and participant feedback collection.

1.8 DISSERTATION STRUCTURE

This dissertation begins with the **Introduction**, which presents the research context, objectives and contributions of this work.

This is followed by the **Related Work** chapter, which introduces the theoretical background and existing research relevant to this study.

The **Proposed Solution** chapter then describes the development of MoodUp and its main components.

Next, the **Methodology** chapter presents the experimental design and evaluation procedure adopted in this research.

The **Results** chapter summarises the findings obtained during the user study.

This is followed by the **Discussion** chapter, which interprets the results and outlines the contributions and limitations of the study.

Finally, the dissertation **ends** with **Conclusions**, a summary of the work and directions for future research.

2. RELATED WORK AND STATE OF THE ART

2.1 STRESS AND EMOTIONAL SELF-REGULATION

2.1.1 STRESS IN MODERN INDOOR ENVIRONMENTS

Mental health and psychological wellbeing are increasingly recognised as important public health challenges. The World Health Organization considers mental health to be an essential component of overall health, yet the demand for mental health support continues to exceed the resources currently available [6]. The World Mental Health Report further highlights that promoting mental wellbeing depends not only on healthcare services but also on creating environments that contribute positively to psychological health.

People are exposed to numerous sources of stress throughout their daily lives, including educational demands, work-related pressures and financial concerns. Because much of everyday life now takes place indoors, researchers have become increasingly interested in understanding how indoor spaces influence emotional experiences and wellbeing. Characteristics of the surrounding environment, including lighting, temperature, air quality and noise, can shape how people feel and experience a space rather than simply determining physical comfort. Consequently, indoor environments are increasingly being viewed as an opportunity to support psychological wellbeing through non-medical interventions [7].

Among the environmental characteristics that may influence wellbeing, temperature, humidity and lighting conditions are considered particularly important. Previous studies have reported environmental ranges that are commonly associated with comfort and positive indoor experiences. A summary of example values identified in the literature is presented in Table 2.1, based on the findings reported in papers [8] and [9].

Table 2.1 – Environmental Parameters Associated with Indoor Wellbeing

Environmental parameter	Ideal range reported	Potential influence
Temperature	22-24℃	Thermal comfort Wellbeing Perceived productivity
Relative humidity	40-60%	Comfort Respiratory system Perceived productivity
Light intensity	200-400 lux	Visual comfort Wellbeing Perceived productivity

Lighting can be considered one of the most influential factors in an environment. Light serves not only a visual function but also plays a critical role in regulating biological processes, including circadian rhythms, sleep patterns and emotional responses. A large-scale population study involving more than 85000 participants found meaningful associations between patterns of light exposure and mental health outcomes [10]. The authors reported that greater exposure to light during the night was associated with an increased risk of several psychiatric conditions, while lighter levels of daytime light exposure were linked to reduced risk of depression, post-traumatic stress disorder and self-harm behaviour. These findings highlight the importance of appropriate lighting conditions and suggest that the environmental light may represent a modifiable factor capable of influencing psychological wellbeing.

Similar findings have also been reported in studies investigating the built environment. Research conducted among active-duty military personnel and veterans found that access to natural light was positively associated with emotional wellbeing and self-reported health outcomes [11]. Participants living in environments with lower levels of natural light reported poorer emotional wellbeing, suggesting that lighting conditions should be taken into account when designing indoor spaces intended to support human health.

The effects of lighting are not limited to residential environments. Recent studies carried out in office settings have shown that both the colour characteristics and dynamic behaviour of lighting systems can influence employee satisfaction and emotional state [12]. In particular, lighting systems designed to mimic natural daylight cycles were associated with lower levels of negative emotions and improved psychological wellbeing when compared with conventional static lighting solutions. These findings suggest that adaptive lighting technologies may provide practical ways of improving indoor environments in workplaces and educational settings.

Taken together, these studies indicate that environmental conditions, and especially lighting, can influence emotional wellbeing in measurable ways. As a result, there is growing interest in developing intelligent systems that can monitor environmental conditions and adjust them in ways that better support users during their daily activities.

2.1.2 PSYCHOLOGICAL MODELS OF EMOTIONAL SELF-REGULATION

Emotional self-regulation describes the way people become aware of, interpret and manage their emotional experiences. Effective self-regulation helps individuals respond to stressful situations, maintain psychological balance and adapt to changes in their surrounding environment. Rather than attempting to eliminate negative emotions entirely, emotional self-regulation focuses on recognising emotional states and adopting strategies that encourage healthy coping and recovery [13].

Contemporary psychological models describe self-regulation as an active and iterative process involving awareness, monitoring, reflection and behavioural adjustment. Recent reviews examining interventions for stress and anxiety management highlight the importance of reflection and self-monitoring as core components of successful self-regulation strategies [14]. Individuals who regularly observe and reflect upon their emotional experiences are generally better equipped to identify stressors, recognise behavioural patterns and implement appropriate coping mechanisms.

Self-regulation is particularly relevant in the context of digital health technologies. Many modern wellbeing applications incorporate mechanisms that encourage users to record their emotional experiences, monitor personal progress and reflect upon changes over time. These practices aim to increase emotional awareness and support the development of healthier long-term habits. Thus, emotional self-regulation has become a central concept within research on digital wellbeing and human-centred technology design.

From an engineering perspective, technologies designed to support self-regulation are not intended to diagnose or treat mental health conditions. Instead, they aim to provide users with tools that encourage emotional awareness, reflection and more informed decision-making. This distinction is particularly important when developing technologies for everyday use, where the objective is generally to support wellbeing and self-reflection rather than deliver clinical interventions.

2.1.3 DIGITAL WELLBEING TECHNOLOGIES

The widespread adoption of smartphones, artificial intelligence and Internet of Things technologies has led to the development of numerous digital wellbeing solutions. Mobile applications, wearable devices and connected environments are increasingly used to support mood tracking, stress management and emotional awareness. These technologies provide users with accessible ways of monitoring different aspects of their daily lives while also offering personalised feedback and recommendations.

Studies investigating mental health applications have identified several features that contribute positively to user engagement, including mood tracking, journaling, progress visualisation and personalised feedback mechanisms [15]. Such features encourage users to take a more active role in managing their wellbeing by presenting emotional information in ways that are easier to understand and reflect upon over time. Similarly, studies conducted among university students have shown that technology-assisted mental health interventions can improve user engagement and encourage self-reflection when usability and accessibility are taken into account [16].

Recent developments are not limited to software-based interventions alone. Researchers have also started looking into systems that combine environmental monitoring, adaptive control mechanisms and emotional information to create more responsive environments capable of supporting user wellbeing. Consequently, smart environments and IoT-based technologies have created new possibilities for bringing together psychological principles and environmental adaptation strategies.

Within this context, adaptive indoor environments have emerged as a promising area of research. By combining emotional self-reporting, environmental sensing and intelligent control mechanisms, these systems may provide personalised experiences that support emotional awareness and wellbeing while remaining accessible to everyday users. The present dissertation follows this research direction by investigating an adaptive lighting system that responds to emotional information extracted from user-generated journal entries.

2.2 JOURNALING AND TEXT-BASED EMOTION RECOGNITION

2.2.1 EXPRESSIVE WRITING AND DIGITAL JOURNALING FOR MOOD TRACKING

Self-reflection is widely regarded as an important process for understanding emotions and maintaining psychological wellbeing. One approach that has received considerable attention is expressive writing, which encourages individuals to put their thoughts and experiences into words. Writing about personal experiences can help people process emotions, recognise recurring patterns and gain a better understanding of the factors that contribute to stress and wellbeing [4].

The increasing use of smartphones and mobile technologies has led to the widespread adoption of digital journaling applications that support regular emotional reflection. In comparison with traditional paper journals, digital platforms often include additional features such as mood tracking, historical visualisation and personalised feedback. Figure 2.1 presents personal examples of reflective digital journal entries illustrating how journaling can be used to organise thoughts and process personal experiences. As a result, journaling has become a common component of many mental health and wellbeing applications.

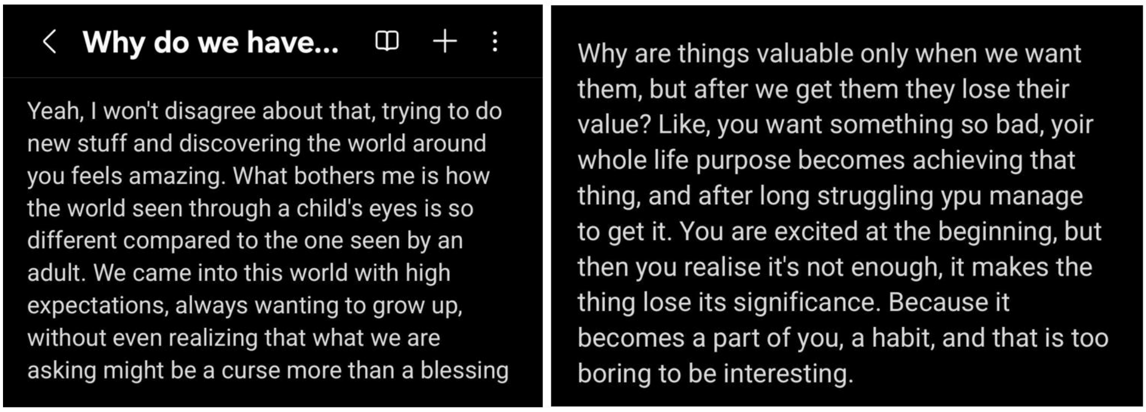

Figure 2.1 - Personal Examples of Journal Entries Used for Emotional Reflection

Research examining user experiences with mental health applications indicates that features related to mood tracking, emotional monitoring and self-reflection are among the most valued by users [17]. A review of mental health screening and diagnostic applications found that users frequently reported positive experiences with tools that enabled them to monitor their emotional state over time and gain a better understanding on their mental wellbeing. These findings suggest that emotional self-reporting may serve not only as a method of data collection but also as a beneficial activity.

Similarly, studies investigating smartphone-based mental health interventions have highlighted the importance of self-monitoring and reflective practices in supporting emotional awareness [15]. Applications that just collect information without encouraging reflection or providing useful feedback often experience low long-term engagement. Successful digital wellbeing systems emphasise user-centred design approaches that promote active involvement rather than passive monitoring.

The present dissertation adopts a journaling-based approach for emotional self-reporting. In contrast to conversational systems that rely on interactions with virtual assistants, journaling encourages users to express their thoughts directly and engage in a process of personal reflection, allowing them to sit with their own thoughts and digest upon them.

2.2.2 NATURAL LANGUAGE PROCESSING AND SENTIMENT ANALYSIS FOR EMOTIONAL DETECTION

The increasing availability of digital text generated through social media and online communication has contributed to the growing interest in NLP techniques for analysing human emotions. NLP encapsulates a range of computational methods designed to process, interpret and extract meaningful information from textual data. Within this field, sentiment analysis has emerged as one of the most widely adopted approaches for identifying emotional tendencies and attitudes expressed through language.

The purpose of sentiment analysis is to classify text content according to its emotional orientation, identifying positive, negative or neutral sentiment. More advanced approaches may also estimate emotional intensity or recognise specific emotions such as happiness, sadness, anger, anxiety, etc. Within mental health and wellbeing research, sentiment analysis has been used to transform written language into measurable information that can be analysed and interpreted [18].

Figure 2.2 illustrates a simplified workflow of text-based sentiment analysis, starting with user-generated text and ending with sentiment classification and system responses.

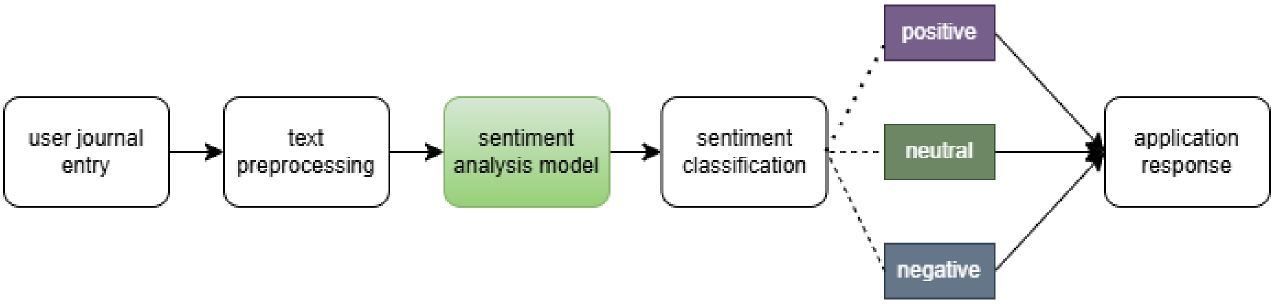

Figure 2.2 - Simplified Sentiment Analysis Process for Emotional Detection from Entry

Recent studies have demonstrated the potential of sentiment analysis as a tool for monitoring emotional states and identifying indicators associated with stress, anxiety and depression [18]. By analysing written language, researchers can gain insight into emotional patterns without relying on intrusive forms of monitoring. This characteristic makes text-based sentiment analysis particularly attractive for applications that prioritise accessibility and user autonomy.

The integration of sentiment analysis into mental health applications has created opportunities for personalised interventions and adaptive feedback mechanisms. Rather than relying solely on questionnaires administered at specific time points, text-based approaches allow emotional information to be collected more continuously through natural user interactions. As a result, sentiment analysis has become an increasingly important component of systems designed to support emotional awareness and self-regulation.

2.3 EFFECTS OF LIGHTING ON HUMAN BEHAVIOUR

2.3.1 LIGHT AND CIRCADIAN RHYTHMS

Light is one of the most influential environmental factors affecting human behaviour, health and wellbeing. Light acts as a biological signal that regulates psychological processes, influences emotional responses and shape the perception of indoor spaces. Recent advances in lighting technology have enabled the development of intelligent system capable of dynamically adapting lighting characteristics to human needs, creating new opportunities for improving comfort, productivity and psychological wellbeing.

Human biological processes are strongly influenced by circadian rhythms, which are approximately 24-hour cycles regulating sleep, hormone production, body temperature, cognitive performance, etc. The primary external factor responsible for synchronising these rhythms is light exposure. Through specialised photoreceptors in the retina, information about environmental lighting conditions is transmitted to the brain, influencing the production of hormones such as melatonin and helping maintain a healthy sleep-wake cycle [19].

Recent research has also pointed out the importance of both light intensity and spectral composition in regulating circadian processes. Exposure to bright, cool light during daytime hours is associated with increased alertness and improved cognitive performance, whereas warmer and dimmer lighting conditions are generally linked to relaxation and preparation for sleep [20]. Thus, lighting conditions that do not align with natural biological rhythms may disrupt circadian processes and have a negative effect on both physical and psychological wellbeing.

The relationship between lighting and mental wellbeing has received growing attention in recent years. A review investigating the effects of artificial light at night reported that excessive exposure to light during evening hours can interfere with melatonin production, reduce sleep quality and contribute to mood disturbances [21]. The authors further highlighted that circadian disruption is one of the main mechanisms through which lighting conditions may influence emotional wellbeing and mental health. Such findings suggest that the design of indoor lighting environments should take into account not only visual comfort but also the biological and psychological responses associated with light exposure.

2.3.2 COLOUR PSYCHOLOGY AND EMOTIONAL RESPONSES

Colour psychology investigates the relationship between colours, lighting characteristics and human emotions. Although emotional responses to colour may differ between individuals and contexts, numerous studies have reported recurring associations between specific colours and particular emotional states.

Research conducted in indoor work environments across multiple countries found that both lighting quality and colour design influenced the psychological mood of participants [22]. Participants reported more positive experiences when lighting conditions were perceived as comfortable and appropriate, whereas excessively dark or overly bright environments were associated with less favourable emotional responses. These findings suggest that the way people perceive lighting conditions may be just as important as objective measurements when designing environments intended to support wellbeing.

Similar findings have also been reported in studies investigating emotional responses to coloured lighting environments. Warm colours such as red, orange and yellow are commonly associated with feelings of warmth, energy and stimulation, while cooler colours such as blue and green are generally linked to calmness, relaxation and emotional stability [23].

Examples of commonly reported colour-emotion associations are presented in Table 2.2. These associations provide a useful theoretical foundation for emotion-aware systems that adapt lighting conditions according to the emotional state of users in an attempt to create more supportive indoor environments [24].

Table 2.2 - Examples of Colours and Associated Emotion

colour	associated emotion
red	energy, agitation, arousal, confidence
orange	warmth, positive, joy, enthusiasm
yellow	joy, happiness, cheerfulness, delight
green	relaxation, calmness
cyan	calmness, peace, safety
blue	tranquillity, soothing
purple	joy, excitement
pink	love, tenderness
grey	fear, sadness
black	depression, sadness

2.3.3 DYNAMIC VERSUS STATIC LIGHTING SYSTEMS

Traditional indoor lighting systems generally rely on fixed illumination settings that remain unchanged regardless of environmental conditions or user needs. Although static lighting can provide adequate visual comfort, it does not adapt to changes in activities, preferences or psychological states throughout the day. The general characteristics of static and dynamic lighting systems are summarised in Table 2.3

Table 2.3 - General Characteristics of Static vs Dynamic Lighting Systems

Characteristic	**Static Lighting**	**Dynamic Lighting**
light intensity	fixed	adjustable
colour temperature	fixed	adaptive
environmental responsiveness	limited	responsive to conditions
personalisation	minimal	higher
potential effect on wellbeing	mainly visual comfort	visual comfort psychological support

More recent developments have shifted attention towards dynamic lighting systems capable of adjusting light intensity, colour temperature, colour composition and temporal patterns over time. Studies indicate that these adaptive approaches may offer several advantages compared with conventional static lighting systems. For example, office workers exposed to dynamic lighting patterns inspired by natural sunlight during stressful tasks reported lower psychological stress responses than participants working under static lighting conditions [25]. They also described the dynamic lighting environment as more natural, engaging and emotionally positive.

Similar findings have also been reported in healthcare settings, where LED systems designed to reproduce natural daylight cycles were associated with improvements in mood and emotional wellbeing [26]. Although the approaches differ, they point in the same direction: adaptive lighting has the potential to provide psychological and physiological benefits beyond those offered by conventional static lighting.

2.4 SMART ENVIRONMENTS AND HUMAN INTERACTION

2.4.1 ENVIRONMENTAL MONITORING FOR WELLBEING

The widespread adoption of low-cost sensors and wireless communication technologies has made it easier to develop smart environments capable of continuously monitoring indoor conditions. Parameters such as temperature, humidity, air quality and lighting levels can now be measured in real time, providing useful information about the environments in which people spend a considerable part of their daily lives.

Researchers have become increasingly interested in understanding how environmental conditions influence psychological wellbeing. Traditional approaches to mental health assessment rely primarily on self-reported information obtained through questionnaires or clinical evaluations. Although these methods remain valuable, they often capture only a limited view of an individual’s day-to-day experiences. As a result, researchers have started looking into ways of combining environmental information with wellbeing monitoring systems. For example, an IoT-based framework integrated environmental measurements with data-driven analysis techniques to investigate factors associated with mental wellbeing [27]. The study suggested that variables such as temperature and humidity may contribute to emotional comfort and should therefore be taken into account when developing systems intended to support psychological health.

The use of environmental monitoring has also been explored outside clinical settings. A recent study involving 84 households evaluated an IoT architecture designed to promote healthier indoor environments through continuous sensing [28]. The system collected information regarding temperature, humidity, carbon dioxide concentration and particulate matter levels and presented these measurements to users through a mobile application. Participants received notifications whenever environmental parameters exceeded recommended thresholds and were encouraged to take corrective actions. Access to environmental information motivated behavioural changes in many households and contributed to measurable improvements in indoor air quality.

Taken together, these studies suggest that environmental monitoring can function not only as a method of data collection but also as a means of increasing user awareness of surrounding conditions. By making environmental information more accessible, smart systems can encourage individuals to make informed decisions about their indoor environments. This principle is particularly relevant to the present dissertation, where environmental measurements are used to provide additional context regarding the conditions of the surrounding environment alongside information about the user's emotional state.

2.4.2 INTERRACTIVE AND ADAPTIVE SMART ENVIRONMENTS

Early IoT systems were primarily designed for monitoring purposes and data collection. More recent developments, however, have increasingly focused on creating environments capable of adapting to user needs and contextual information. These systems move beyond passive observation by incorporating mechanisms that automatically adjust environmental conditions or allow users to influence system behaviour through interaction. As a result, human-centred smart environments have gained considerable attention, with technology being developed to support users in more personalised and meaningful ways.

One of the defining characteristics of adaptive environments is the presence of a continuous feedback loop between the user and the system. Information gathered from sensors is processed to identify relevant patterns and is subsequently used to influence system behaviour. Research investigating environmental monitoring within mental health applications has shown that combining environmental measurements with user-specific information can provide a broader understanding of wellbeing than either source independently [27]. Rather than relying on fixed configurations, these approaches aim to create systems that can respond to individual circumstances as they change.

Researchers have also started looking into the use of contextual information for personalising digital health interventions. For example, studies investigating the integration of IoT technologies and advanced artificial intelligence techniques have proposed architectures capable of adapting system behaviour according to user context and environmental conditions [29]. Although many of these approaches incorporate sophisticated machine learning and large language model technologies, the underlying principle remains straightforward: smart environments become more effective when they can adjust their behaviour according to the needs of individual users.

Adaptation is particularly important in environments intended to support wellbeing. Human emotions are dynamic and continuously influenced by everyday experiences and external events. Static systems are often unable to provide appropriate support across different situations because they cannot react to these changes. Adaptive environments attempt to overcome this limitation by continuously responding to relevant information and modifying environmental conditions accordingly.

2.5 EMOTIONAL-AWARE LIGHTING SYSTEMS

2.5.1 PRIOR ACADEMIC PROTOTYPES AND FRAMEWORKS

Recent advances in human-centred computing have led to the development of intelligent systems capable of adapting environmental conditions according to the emotional state of users. Researchers have explored various approaches for detecting emotions and translating them into environmental changes, particularly through smart lighting technologies. These systems generally combine emotion recognition techniques with IoT infrastructures in an attempt to improve comfort, emotional awareness and wellbeing.

One relevant example is the emotional lighting system proposed by Cho et al. [30], which analyses instant message conversations to identify the emotional state of occupants and automatically adjust indoor lighting conditions. The system extracts textual information from messaging applications, performs sentiment analysis using IBM Watson Natural Language Understanding and communicates lighting commands to an IoT-enabled device. The authors demonstrated that emotional information obtained from written communication can be transformed into adaptive lighting responses that create a more personalised indoor environment. While the proposed architecture shares similarities with the system presented in this dissertation, it relies heavily on cloud-based processing and passive collection of private messaging data.

A different perspective was provided by MacIsaac et al. [4], who investigated the effects of a smartphone-based journaling intervention on psychological wellbeing and self-reflection. Their longitudinal study involving 152 participants demonstrated that app-based journaling activities were associated with improvements in psychological wellbeing and highlighted the importance of self-reflective processes during digital writing activities. Unlike the present dissertation, the study did not incorporate environmental adaptation mechanisms or emotion-aware responses capable of modifying the user's surroundings. Nevertheless, the findings provide empirical support for the use of digital journaling as a tool for promoting emotional reflection and wellbeing.

Emotion-aware adaptation has also been investigated in smart home environments. Yamauchi et al. [31], proposed an automation framework based on the emotional Biologically Inspired Cognitive Architecture ('eBICA'), which adapts environmental behaviour according to user emotions. Their proof-of-concept experiment demonstrated reductions in state anxiety following emotion-based automation and highlighted the potential of personalised adaptive environments for promoting psychological wellbeing. Unlike the present dissertation, the proposed framework focused primarily on automation strategies and anxiety reduction rather than journaling and user perceptions of adaptive lighting.

In addition, Lorincz et al. [32] investigated the application of usability testing and the System Usability Scale in the evaluation of digital systems. Although the study did not focus on emotion-aware environments, it demonstrated the usefulness of SUS as a reliable instrument for assessing perceived usability during system development and validation. The study therefore provides methodological support for the evaluation strategy adopted in the present dissertation.

Taken together, these studies demonstrate growing interest in emotion-aware systems that combine emotional information, digital technologies and adaptive behaviours to support user wellbeing. Nevertheless, existing studies tend to investigate adaptive lighting, digital journaling, emotion-aware automation and usability evaluation as separate research directions. Comparatively fewer studies examine the integration of reflective journaling, sentiment analysis, environmental monitoring and adaptive lighting responses within a single user-centred platform.

2.5.2 ANALYSIS OF CURRENT COMMERCIAL SOLUTIONS

A. *Digital Journaling and Mood-Tracking Applications*

In parallel with academic research, numerous commercial applications have emerged to support emotional wellbeing through self-reflection, mood tracking and journaling activities. Many of these applications encourage users to record emotions regularly and provide visual summaries intended to promote emotional awareness and long-term self-monitoring. Figure 2.3 presents examples of popular commercial applications.

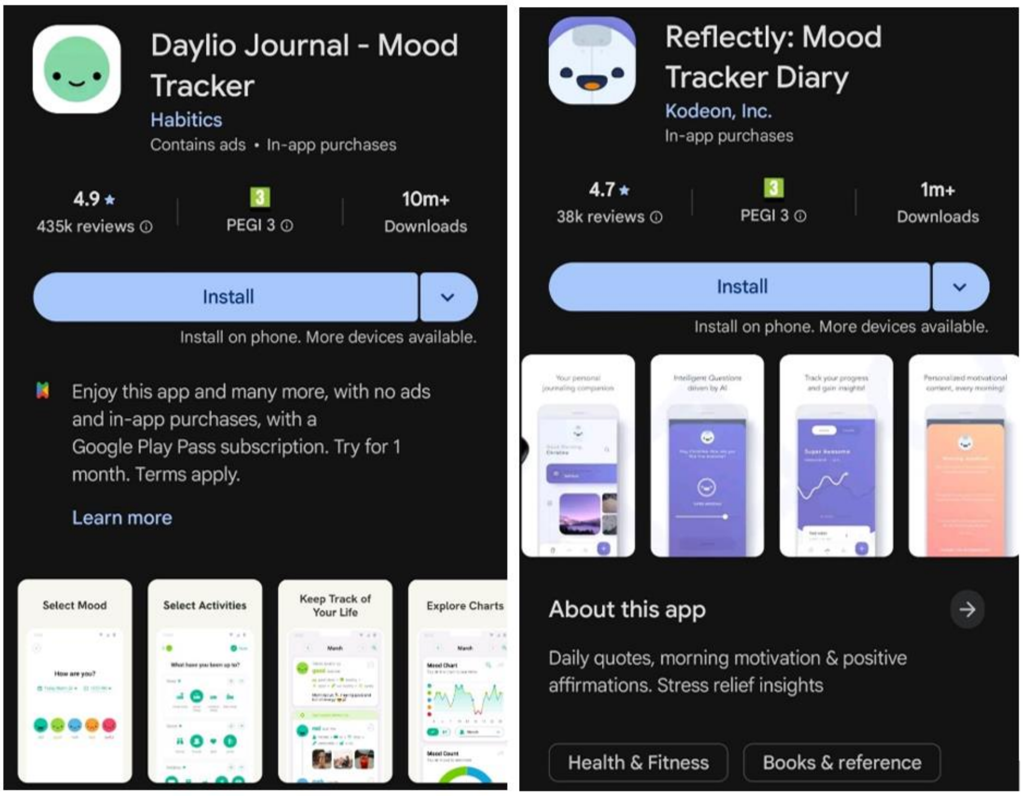

Figure 2.3 – Examples of Commercial Digital Journaling and Mood Tracking Applications

Daylio [33] is a mood-tracking application that allows users to record emotions, activities and personal notes through a simple and highly visual interface. The application generates statistics and historical charts that help users identify emotional patterns and behavioural trends over time.

Reflectly [34] combines digital journaling with artificial intelligence techniques to encourage self-reflection through guided questions and mood tracking features. The application places particular emphasis on emotional awareness by prompting users to describe their experiences and review their emotional progress through personalised visualisations

Despite these advantages, commercial journaling applications remain largely confined to the digital domain. Although they effectively collect emotional information and provide analytical feedback, they rarely influence the physical environment surrounding the user. Consequently, emotional reflection and environmental adaptation remain separate processes, requiring users to translate emotional insights into behavioural changes manually.

B. *Smart Lighting Ecosystems and Wellness Hardware*

The commercial smart lighting market has expanded considerably in recent years, providing users with the ability to control colour, brightness and lighting schedules through mobile applications, voice assistants and automation platforms. Examples of widely adopted smart lighting ecosystems are presented in Figure 2.4.

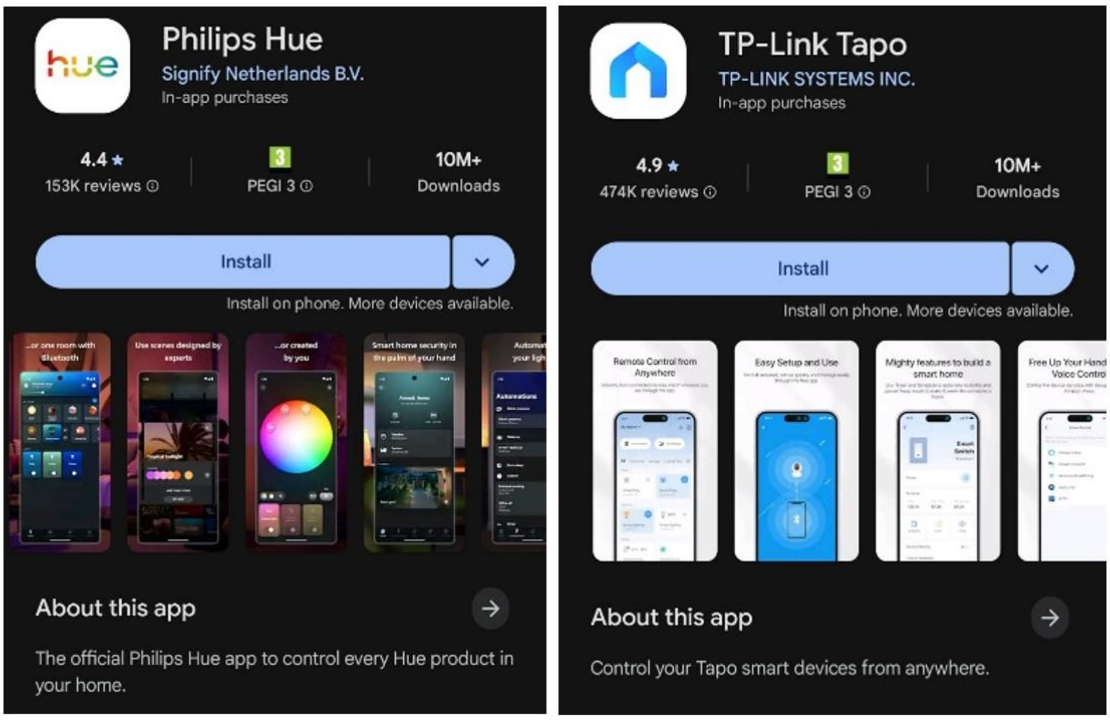

Figure 2.4 – Examples of Commercial Smart Lighting Applications

Philips Hue [35] is one of the most established smart lighting ecosystems and offers extensive personalisation capabilities, including dynamic scenes, colour control, automation routines and integration with third-party services. The platform is designed to support both functional and ambient lighting experiences within residential environments

Tp-Link Tapo [36] provides a more accessible smart lighting platform that enables users to modify brightness, colour and scheduling parameters through a mobile application. Similar to ‘Philips Hue’, the platform supports automation scenarios and remote control features while maintaining a relatively simple user experience

However, current commercial products generally depend on manual user interaction or predefined automation rules. Lighting adjustments are typically triggered by schedules, occupancy sensors or user commands rather than by an understanding of the emotional state of users. While these platforms provide extensive control over environmental conditions, they lack mechanisms for integrating personal emotional information into the adaptation process.

2.6 RESEARCH GAP AND POSITIONING OF THE PROPOSED SYSTEM

2.6.1 LIMITATIONS OF EXISTNG APPROACHES

The literature reviewed in the previous sections demonstrates significant progress in the fields of digital wellbeing, emotion recognition, smart lighting, and human-centred IoT systems. Regardless, several limitations remain evident across both academic research and commercially available solutions.

Academic prototypes have successfully demonstrated the utility of adapting indoor environments according to the emotional states of the users. However, many of these approaches rely on cloud-based processing infrastructure, requiring the transmission of personal information to external services. Such architectures introduce concerns regarding privacy, data security and long-term scalability. Furthermore, emotional information is often obtained through passive sources such as instant messaging data, camera feeds or physiological sensors. While effective, these approaches may reduce user awareness and engagement in the emotional reflection process as they may be perceived as intrusive.

Commercial wellbeing applications face different limitations. Journaling and mood-tracking platforms provide valuable tools for self-reflection, emotional awareness and long-term monitoring. However, these applications generally remain limited to the digital environment and do not influence the physical surroundings to the user. Emotional insights are visualised through statistics, charts or reports, but the responsibility for translating these insights into behavioural or environmental changes remains entirely with the user.

Similarly, commercial smart lighting ecosystems provide extensive control over environmental conditions through mobile applications, automation routines and scheduling features. Despite their versatility, these systems usually operate independently of the emotional state of the user. Lighting adjustments are commonly triggered by predefined rules, occupancy detection or manual interaction rather than by emotional context.

Taken together, current solutions tend to address emotional monitoring and environmental adaptation as separate challenges. Comparatively few systems provide a unified framework that combines emotional self-reflection, sentiment analysis, environmental awareness and adaptive lighting control within a single user-centred platform.

2.6.2 THE IDENTIFIED RESEARCH GAP

The literature and commercial solutions reviewed throughout this chapter indicate that a gap still exists between emotional self-reflection and adaptive smart environments. Previous studies have proposed emotion-aware lighting systems; however, these solutions often rely on passive methods of collecting emotional information and frequently provide limited opportunities for users to actively reflect on their emotions. Conversely, journaling and mood-tracking applications encourage self-reflection and emotional awareness but generally lack the ability to influence the physical environment surrounding the user.

Similarly, many commercial smart lighting platforms have been designed primarily to improve convenience, energy efficiency and visual comfort, while giving relatively little attention to emotional wellbeing. Although increasing evidence points towards a relationship between environmental conditions and mental health, there remains a limited number of integrated systems capable of translating reflective emotional information into meaningful environmental adaptation.

The present dissertation was developed in response to this gap by bringing together several complementary functionalities within a single platform. The proposed system allows users to record their thoughts through digital journaling, analyses the emotional content of journal entries using local sentiment analysis techniques and adapts ambient lighting according to the detected emotional tone. In addition, environmental parameters such as temperature, humidity and ambient light levels are continuously monitored and presented through a mobile application, providing users with additional awareness of their surroundings.

Rather than treating emotional monitoring and environmental control as separate processes, the proposed system establishes a direct connection between emotional reflection and adaptive environmental feedback. This integration forms the central contribution and research focus of the present dissertation.

2.6.3 MAIN CONTRIBUTIONS OF THIS DISSERTATION

Based on the research gap identified in the previous section, this dissertation makes several contributions within the fields of smart environments, Internet of Things technologies and digital wellbeing.

The first contribution is the development of a mobile application that supports digital journaling and emotional self-reflection. The application enables users to record experiences as they occur and observe emotional trends over time through personalised visual summaries.

The second contribution consists of a local sentiment analysis framework capable of classifying emotional content directly on the user's device. By performing sentiment analysis locally rather than through cloud services, the proposed approach reduces dependence on external platforms and allows the system to remain operational regardless of network connectivity.

The third contribution involves the implementation of an IoT-based ambient lighting device that adapts lighting conditions according to the emotional information detected from journal entries. The system combines sentiment analysis with colour-based lighting responses inspired by previous research on colour psychology and human-centric lighting.

The fourth contribution is the integration of environmental monitoring capabilities, including temperature, humidity and ambient light sensing. These measurements provide additional contextual information regarding the surrounding environment and contribute to the development of a more comprehensive smart environment.

The last contribution is the experimental evaluation of the proposed system through a user study based on questionnaire-based assessment methods. Unlike the author's previous bachelor's project, which primarily focused on system development and technical feasibility, the present work investigates user perceptions of adaptive lighting responses and evaluates the system's usability and potential to support emotional awareness and self-reflection.

Through these contributions, this dissertation proposes a human-centred platform that combines reflective digital practices, environmental monitoring and adaptive lighting technologies to support emotional awareness and wellbeing.

2.6.4 SYSTEMS FEATURES COMPARISON

To better position the proposed system within the existing research and commercial landscape, a comparative analysis of representative academic prototypes and commercial solutions is presented in this section. The comparison focuses on characteristics relevant to the objectives of the present dissertation, including emotion recognition approaches, environmental adaptation capabilities, user interaction mechanisms, sentiment analysis techniques and evaluation methodologies.

Table 2.4 compares MoodUp with the representative academic prototypes presented in Subchapter 2.5.1. The comparison shows that existing systems typically address only individual aspects of emotion-aware environments, whereas MoodUp integrates multiple functionalities within a single platform and additionally incorporates environmental monitoring and user-centred evaluation.

Table 2.4 – Comparison of MoodUp with Other Academic Prototypes

Feature	Cho et al.	MacIsaac et al.	eBICA Framework	MoodUp (dissertation project)
digital journaling	no	yes	no	yes
sentiment or emotion recognition	yes	no	partial	yes
adaptive lighting	yes	no	no	yes
emotion-aware adaptation	yes	no	yes	yes
environmental monitoring	no	no	partial	yes
psychological wellbeing focus	partial	yes	yes	yes
user evaluation study	partial	yes	yes	yes
longitudinal evaluation	no	yes	no	no
usability evaluation (SUS)	no	no	no	yes

Table 2.5 compares MoodUp with commercially available applications and smart lighting ecosystems. Since no commercial solution was identified that combines emotional self-monitoring and adaptive lighting within a single platform, the comparison is divided between representative journaling applications and smart lighting products. By contrast, MoodUp integrates both approaches within a unified system designed to support emotional awareness and wellbeing.

Table 2.5 – Comparison of MoodUp with Other Products on the Market

Feature	**Dailyo**	**Reflectly**	**Philips Hue**	**Tp-Link Tapo**	**MoodUp (dissertation project)**
digital journaling	yes	yes	no	no	yes
mood tracking	yes	yes	no	no	yes
sentiment analysis	no	limited AI guidance	no	no	yes
environmental monitoring	no	no	no	no	yes
adaptive lighting	no	no	yes	yes	yes
emotion-aware adaptation	no	no	no	no	yes
personalised feedback	limited	yes	no	no	yes

3. PROPOSED SOLUTION

3.1 SYSTEM ARCHITECTURE OVERVIEW

Figure 3.1 presents the overall system architecture and the interactions between the user, the mobile application, the cloud infrastructure and the IoT device.

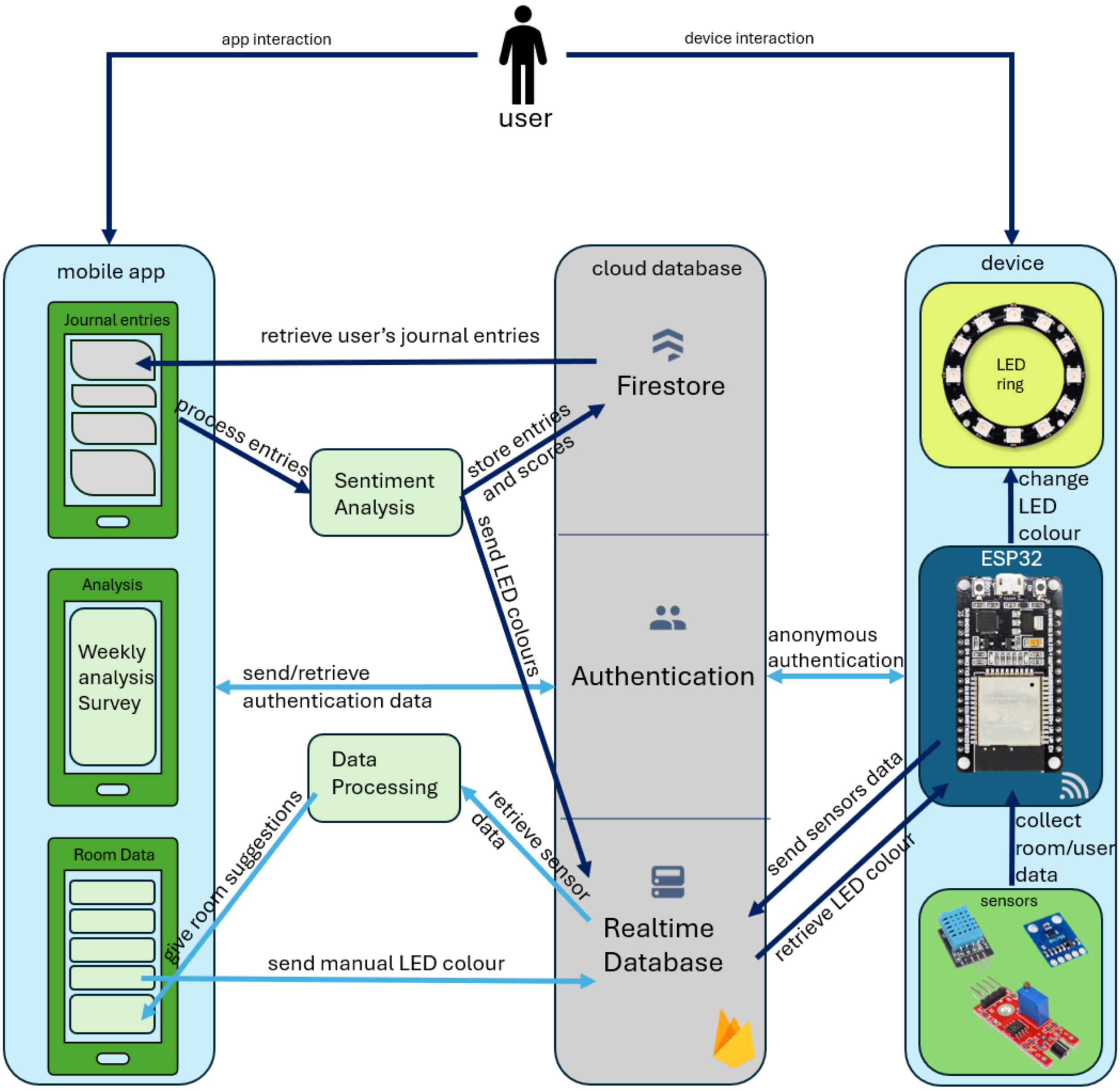

Figure 3.1 – System Architecture of MoodUp

MoodUp consists of three principal components: an Android mobile application, a cloud infrastructure implemented using Firebase services and an ESP32-based IoT device equipped with an RGB LED ring and environmental sensors. Together, these components make up an adaptive ambient environment capable of responding to emotional information extracted from user journal entries.

From the user's perspective, interaction with the system begins by turning on the IoT device and signing in to the mobile application. Here, the user can then record journal entries, monitor environmental conditions and review emotional information collected over time. After the emotional tone is detected, the device changes the colour of the light. In addition, the user can manually modify the lighting when desired and receive simple suggestions based on the monitored room conditions.

While these interactions are visible to the user, data storage and communication happen in the background. Firebase services are in charge for user authentication, data persistence and real-time communication between the app and the IoT device, allowing all components of the system to remain synchronised.

Compared with the previous undergraduate implementation of MoodUp, the current system introduces several additional capabilities, including structured digital journaling, local sentiment analysis, emotional trend visualisation, environment suggestions and a formal user evaluation study. These additions extend the original prototype into a more thorough emotion-aware smart environment. More implementation details regarding the original MoodUp prototype can be found in the author's Bachelor's Thesis [5]. The current implementation version and more details can be found in the author's dissertation repo [37].

3.2 MOBILE APPLICATION

The mobile application represents the main interface that allows users to interact with the MoodUp system. It was developed for the Android platform using Kotlin and follows the Model-View-ViewModel (MVVM) architectural pattern. This approach separates the user interface from business logic and data management components, improving maintainability and simplifying future development.

The application is organised around three principal sections that support the core functionalities of MoodUp, including emotional journaling, environmental monitoring and emotional history visualisation. Data persistence and user management are supported through Firebase services, while communication with the IoT device is achieved through Firebase Realtime Database.

Figure 3.2 provides an overview of the principal screens and internal architecture of the MoodUp mobile application.

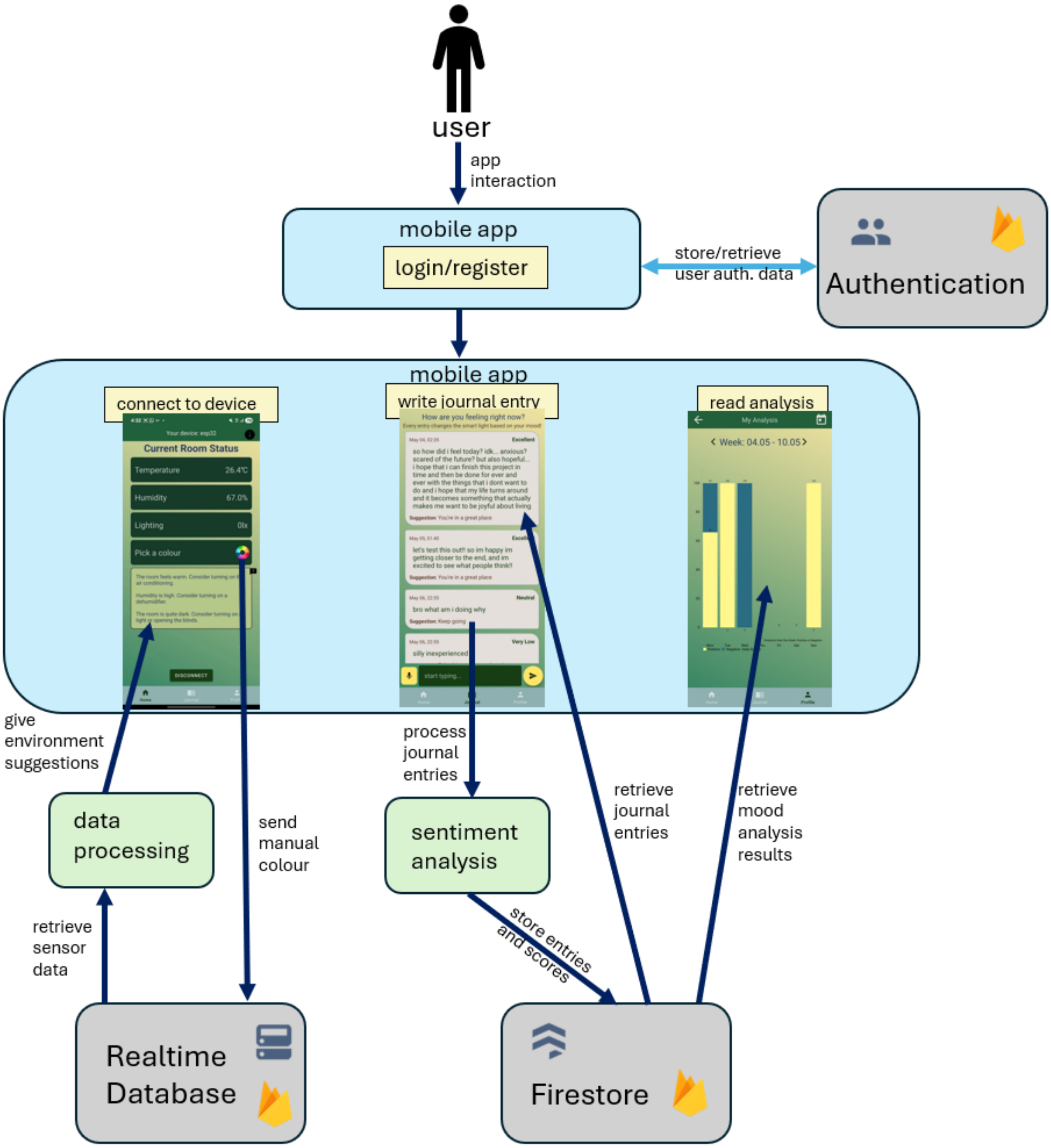

Figure 3.2 – Main Screens and Internal Architecture of MoodUp's Mobile Application

3.2.1 USER MANAGEMENT AND AUTHENTICATION

User management is implemented through Firebase Authentication and provides registration and login functionality for the MoodUp application. Authentication is used to associate journal entries, sentiment results and user preferences with individual accounts, ensuring that personal information remains isolated between users.

After successful authentication, users gain access to the main application components managed by the MainActivity class and navigated through a BottomNavigationView. The application is organised into three principal fragments: HomeFragment, JournalFragment and ProfileFragment.

The use of authenticated accounts also supports the privacy requirements of the system. Journal entries and emotional information are linked to the currently authenticated user rather than being stored as shared application data, allowing each user to access only their own emotional history and environmental information.

3.2.2 DIGITAL JOURNALING MODULE

The digital journaling module forms one of the main components of the MoodUp system. Through the JournalFragment, users can record short entries describing recent experiences through felt emotions. The objective of this module is to encourage self-reflection while providing the textual input required for sentiment analysis. Figure 3.3 presents an example of a journal entry after sentiment analysis and the personalised feedback message displayed to the user.

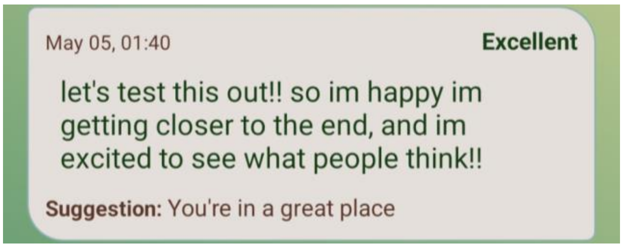

Figure 3.3 - Example of a Journal Entry and Generated Feedback

Once a journal entry is submitted, it is processed by the sentiment analysis module and assigned a sentiment classification. The application then displays a predefined feedback message intended to provide a more engaging and personalised journaling experience.

Journal entries, sentiment results and timestamps are stored in Firebase Firestore. This information can later be retrieved to generate visual summaries of emotional patterns and enable users to review previous entries. As a result, the journaling module supports both immediate interaction with the adaptive lighting system and longer-term emotional self-monitoring.

3.2.3 SENTIMENT ANALYSIS MODULE

The sentiment analysis module is responsible for determining the emotional orientation of journal entries submitted by users. In the current implementation, sentiment analysis is performed locally within the mobile application using the AFINN sentiment lexicon. This approach eliminates the need to transmit journal content to external processing services and allows emotional analysis to be performed directly on the user's device.

After a journal entry is submitted, the text is tokenised and individual words are compared against the AFINN dictionary [38], where each term is associated with a predefined sentiment score. Figure 3.4 presents examples of words and associated scores contained within the AFINN sentiment lexicon.

helpless	-2
helps	2
hero	2
heroes	2
heroic	3
hesitant	-2
hesitate	-2
hid	-1
hide	-1

salute	2
saluted	2
salutes	2
saluting	2
salvation	2
sappy	-1
sarcastic	-2
satisfied	2

Figure 3.4 – Example of Words and Associated Scores from the AFINN Lexicon

Individual word scores are aggregated to produce an overall sentiment value representing the emotional tone of the journal entry.

The resulting sentiment information serves two purposes. First, it is stored as part of the user's emotional history and later used for visualisation and trend analysis. Second, the sentiment result is transmitted through Firebase services to the IoT device, which updates the colour of the ambient lighting according to the detected emotional state.

Compared with the previous implementation of MoodUp, the use of local sentiment analysis reduces dependence on external services, improves portability and provides a more privacy-oriented approach to processing personal journal content.

3.2.4 VISUALISATION AND ANALYTICS

The application includes visualisation features that enable users to review their emotional history over time. Through the ProfileFragment, users can access a mood analysis section where previous sentiment results are displayed visually. This allows users to observe patterns in their journal entries, such as the number of positive and negative entries recorded across different days.

The HomeFragment displays environmental data collected by the IoT device, including room temperature, humidity and light intensity. The same section also allows users to view room-related suggestions based on the monitored conditions and manually adjust the lighting when needed.

Together, these features allow MoodUp to provide immediate feedback through adaptive lighting while also supporting longer-term emotional self-monitoring with visualisation and environmental awareness.

3.3 IOT DEVICE

Figure 3.5 presents the hardware architecture of the proposed IoT device.

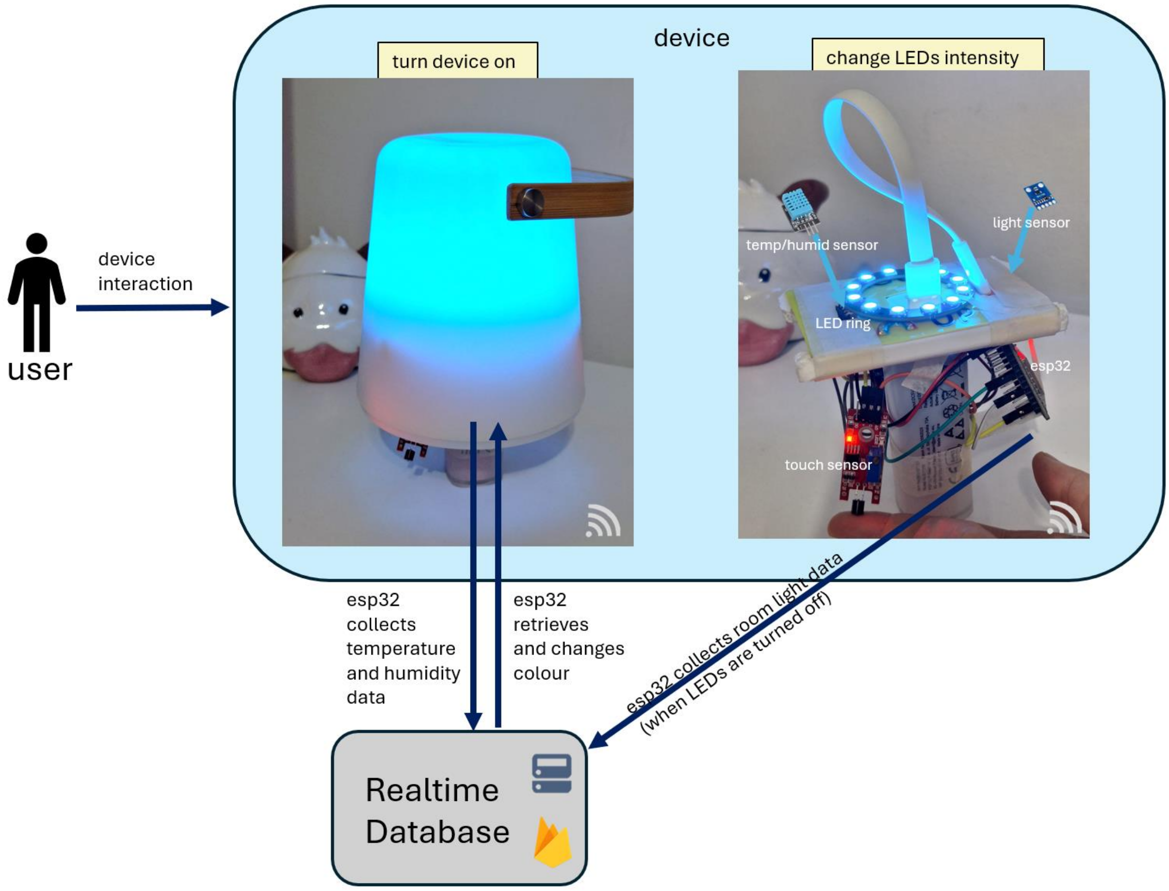

Figure 3.5 – Hardware Architecture and Principal Components of the MoodUp IoT Device

The IoT device represents the physical component of the MoodUp system and is responsible for adaptive lighting control and environmental data collection. The device was developed around an ESP32 microcontroller and integrates a 12-LED WS2812B RGB ring, a DHT11 temperature and humidity sensor, a GY-302 BH1750 ambient light sensor and a TTP223 capacitive touch sensor module within a single platform.

The device firmware was implemented in C++ using the Arduino Integrated Development Environment (IDE). The ESP32 communicates with Firebase Realtime Database to receive lighting information generated by the mobile application and to transmit environmental measurements collected from the surrounding environment. Through this communication process, the device forms the link between digital user interactions and the surrounding physical environment.

3.3.1 SMART LIGHTING CONTROL SYSTEM

The smart lighting subsystem provides visual feedback based on the emotional information detected within user journal entries. The lighting system is implemented using a 12-pixel WS2812B RGB LED ring controlled by the ESP32 microcontroller.

After a journal entry is analysed within the mobile application, the resulting sentiment value is transmitted through Firebase Realtime Database. The ESP32 periodically retrieves this information and updates the colour of the LED ring accordingly, allowing the surrounding lighting environment to respond automatically to the emotional content of the journal entry.

The proposed system uses a predefined sentiment-to-colour mapping inspired by findings from colour psychology and human-centric lighting research, as illustrated in Figure 3.6. Cooler colours, including blue and cyan, are associated with lower emotional states and are intended to create a calming atmosphere. As the detected sentiment becomes more positive, the lighting gradually transitions towards green, yellow and magenta hues, which are associated with increasingly positive emotional states.

Figure 3.6 – Sentiment-to-Colour Mapping Implemented in MoodUp

In addition to automatic operation, the lighting subsystem supports manual control through the mobile application and local brightness adjustment using the capacitive touch sensor integrated into the device. This combination of automatic adaptation and manual interaction provides additional flexibility while preserving the emotion-aware behaviour of the system.

3.3.2 ENVIRONMENTAL MONITORING SYSTEM

The environmental monitoring subsystem was developed to collect contextual information regarding the indoor environment in which the system operates. The device incorporates a DHT11 sensor for measuring temperature and humidity and a GY-302 BH1750 sensor for measuring ambient light intensity.

Sensor measurements are periodically collected by the ESP32 and transmitted to Firebase Realtime Database. The mobile application retrieves these values and presents them to users in real time together with simple room-related recommendations derived from the recorded measurements.

Although environmental parameters were not considered main evaluation variables during the user study, they represent an important extension of the proposed platform. They are used to provide additional information to the user based on which they get environmental suggestions that might help to further improve their wellbeing.

3.3.3 LOCAL DEVICE LOGIC

Figure 3.7 illustrates the simplified operational workflow executed by the ESP32 firmware during system operation.

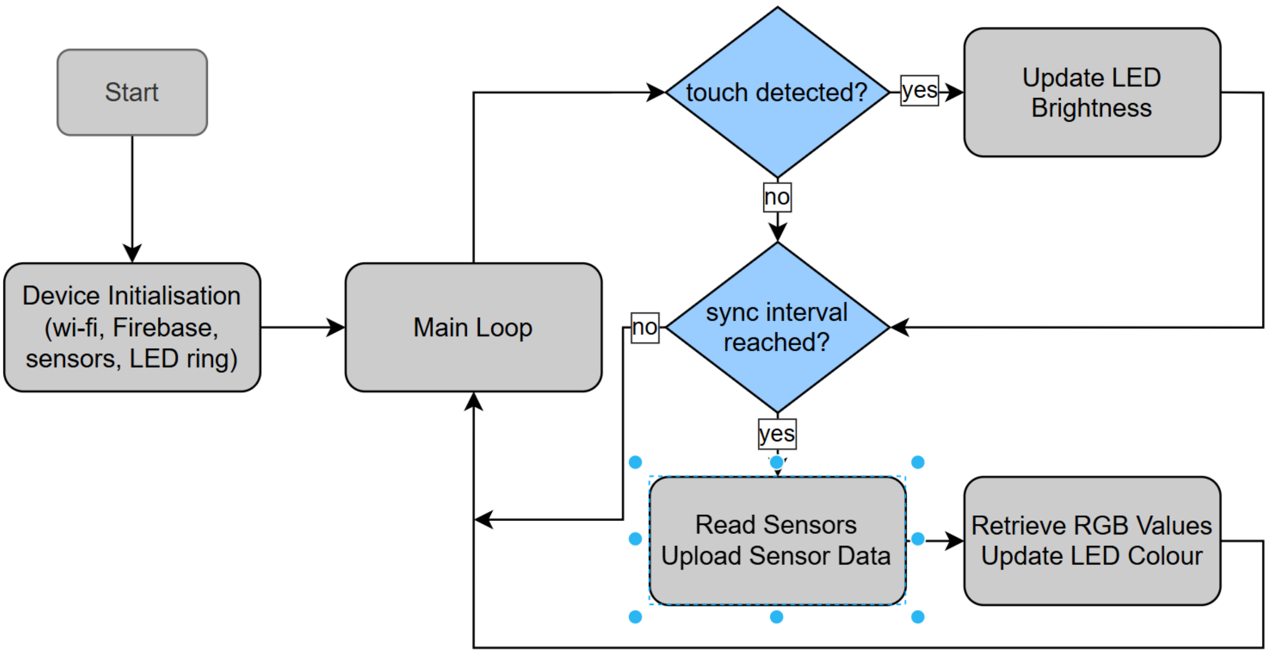

Figure 3.7 - Flowchart of the Local Device Logic Executed by the ESP32

The ESP32 executes a continuous control loop responsible for communication management, environmental sensing and lighting control. During operation, the microcontroller continuously monitors the state of the capacitive touch sensor and periodically synchronises data with Firebase Realtime Database.

At regular intervals, the device acquires temperature, humidity and ambient light measurements and uploads these values to Firebase. The ESP32 also retrieves the latest RGB values generated by the mobile application and updates the LED ring accordingly, allowing the ambient lighting to respond automatically to the emotional information derived from user journal entries.

The combination of cloud communication, environmental monitoring and local user interaction allows the device to operate as an adaptive and responsive component within the overall MoodUp architecture.

3.4 CLOUD INFRASTRUCTURE

The cloud infrastructure of MoodUp was implemented using Firebase services. Firebase Authentication, Cloud Firestore and Firebase Realtime Database were integrated to support user management, data persistence and communication between the IoT device and mobile app. This approach allows information to be exchanged between software and hardware components without requiring a direct connection between them.

Authentication is handled differently for the mobile application and the IoT device. The mobile application uses Firebase Authentication to create user accounts and associate journal entries, sentiment results and mood history information with individual users.

The ESP32 device performs anonymous authentication in order to access Firebase Realtime Database and exchange sensor measurements and lighting information. This approach simplifies device deployment, as the hardware component can communicate with Firebase without requiring separate visible user credentials.

Cloud Firestore is used to store information that must remain available over time. The database follows a NoSQL document-based structure, making it suitable for storing journal entries and calculated mood information as flexible documents rather than fixed relational tables.

In MoodUp, each authenticated user is represented as a document inside the journalUsers collection. Each user document contains two principal subcollections. The journalEntries subcollection stores individual journal entries and associated sentiment information, while the dailyScores subcollection stores calculated daily positive and negative percentages used for mood visualisation.

The simplified Firestore structure that the application is based on can be seen in Figure 3.8.

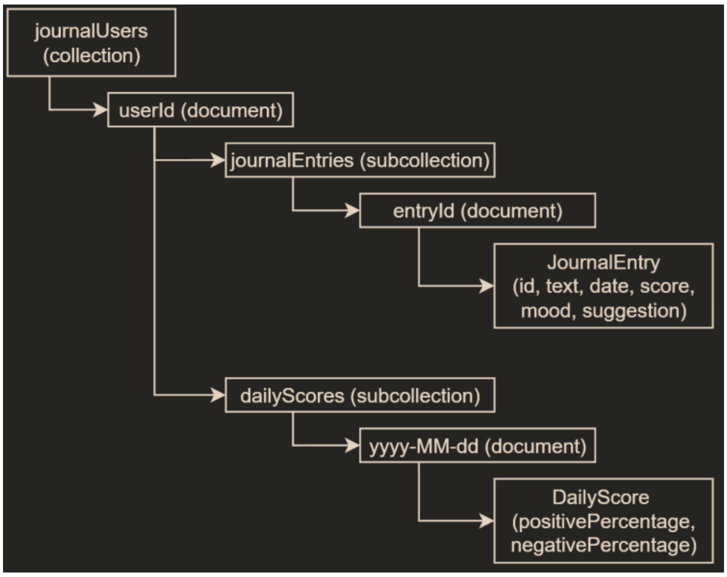

Figure 3.8 - Example of the Cloud Firestore Structure Used by MoodUp

Firebase Realtime Database is used for information that requires more frequent synchronisation between the mobile application and the IoT device. This includes the RGB colour values retrieved by the ESP32 and the environmental measurements uploaded by the hardware component.

Figure 3.9 presents an example of the Realtime Database structure used during system operation.

Figure 3.9 - Example of the Firebase Realtime Database Structure Used by MoodUp

The Realtime Database structure contains a root node named devices that stores information for each registered ESP32 device. Each device node contains three principal branches responsible for lighting information, user association and environmental measurements. The lighting branch stores the red, green and blue intensity values used by the RGB LED ring, the user branch stores the identifier of the user currently associated with the device, and the sensor branch stores temperature, humidity and ambient light measurements collected by the ESP32.

By separating persistent journal data from real-time device information, MoodUp uses each Firebase service according to its intended purpose. Cloud Firestore supports the long-term storage of user and mood information, while Firebase Realtime Database enables lightweight synchronisation between the mobile application and the ESP32 device.

4. METHODOLOGY

4.1 STUDY DESIGN

In order to evaluate MoodUp, which is an adaptive ambient lighting system that responds to the sentiment expressed through digital journal entries, a user study was conducted. The scope of this study was to see how the participants experienced MoodUp and to see if the adaptive lighting condition was perceived more positively than a traditional room lighting condition.

To achieve this, the chosen approach was to use a within-subject design. For MoodUp, this entailed that every participant tested the system under two different conditions: one using the existing lighting conditions available in the room, the other using the previously developed lighting system. The study's centre point was that the same people tested both situations to remove personal bias, as each participant had their own baseline for comparison.

As mentioned above, the experiment consisted of two different scenarios both carried out during the same testing session. In the first scenario, **Scenario A,** participants interacted with the mobile app while the lighting conditions of the room remained unchanged regardless of the content of their journal entry. This was the baseline environment, the static one, against which the proposed system was evaluated.

In the second scenario, **Scenario B**, participants used the same application while the proposed adaptive lighting system was up and running. The sentiment analysis component processed the submitted journal entry and automatically adjusted the colour of the light (based on the detected emotional tone). This was the intervention environment, the adaptive one, which demonstrated the main functionalities of MoodUp.

As the scenarios unfolded, participants completed a short questionnaire that was the same for both scenarios, to evaluate their perception to the environment they were in. After the scenarios were completed, there were a few more questions about the usability of the developed system and a small demographic quiz.

The purpose of the comparison between the two scenarios was to see if an adaptive lighting environment would be perceived as more supportive and responsive than a traditional room lighting environment by the participants. The study was also designed to show if participants considered MoodUp to be a useful tool for emotional self-regulation while remaining easy to use.

4.2 PARTICIPANTS

MoodUp's evaluation study involved a total of N = 17 participants. They were recruited through a sampling approach, selected from the author's personal and academic contacts, including peers, friends, family members, etc. Taking into account the exploratory nature of the study, this strategy was considered appropriate. The objective was to evaluate a functional prototype rather than establishing population-wide conclusions.

To better understand the participants, a demographic questionnaire was added at the end of the evaluation. Information regarding age range, gender, and other various questions related to the participant's knowledge and interest in adaptive lighting systems was collected. This information was collected to better describe the participant group and support the interpreted results.

The only inclusion criterion for the participants was the ability to use a smartphone application on their own. There were no requirements related to previous mental health diagnoses or prior experience with journaling. MoodUp's goal is to support emotional reflection and wellbeing in the general population rather than a specific clinical group.

The main goal of this study was to evaluate the perception of participants about MoodUp in comparison with existing room lighting conditions, so the participant sample was considered enough for this initial proof-of-concept evaluation.

4.3 EXPERIMENTAL PROCEDURE

For each participant, the experimental procedure lasted around 15-20 minutes. The study took place inside a quiet indoor environment with a functioning light source (already available in the room).

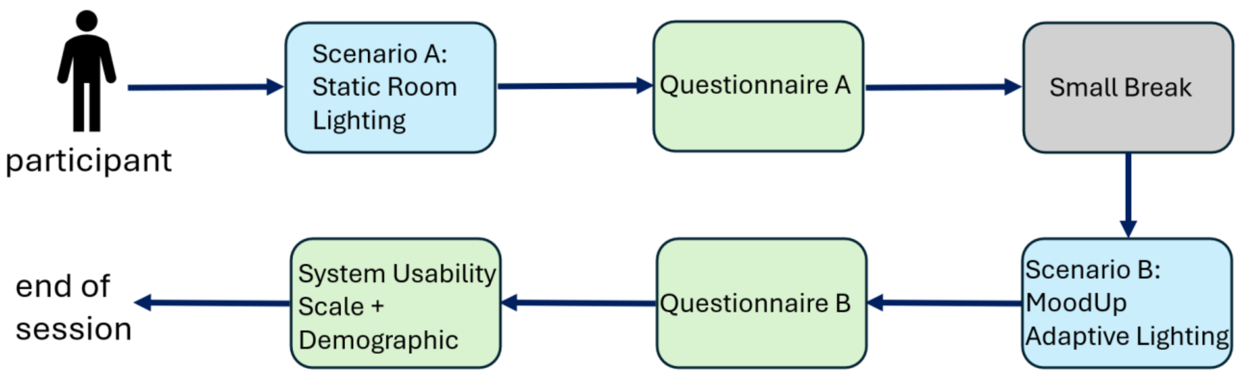

Figure 4.1 - Overview of the Evaluation Procedure

Figure 4.1 presents a step-by-step overview of the evaluation procedure followed during the study.

The only equipment required was a mobile phone with the application installed and the MoodUp lighting device.

At the start of the session, each participant was introduced to MoodUp and told what it does. They were informed about the purpose of the study: to evaluate user interaction with a traditional lighting environment versus an adaptive ambient lighting system. They did not require to have any specialised technical knowledge.

The experiment consisted of two separate testing conditions. **In the first condition**, Scenario A, the participant interacted with the mobile app while the MoodUp lighting device remained inactive. They were asked to write a short journal entry describing a recent emotional experience, focusing on their felt emotions, rather than on the event itself. During this phase, the lighting environment remained unchanged regardless of the content of the journal entry. After completing the journaling task, participants filled in the first section of the evaluation questionnaire.

Between the two conditions, the participant took a short break of approximately two minutes. This was to reduce the influence of the previous task and allow the participant to refocus before continuing.

In the second condition, Scenario B, the participant was asked to write a new journal entry describing a different emotional experience. During this phase, MoodUp processed the text using the sentiment analysis component. Then, it automatically adjusted the lighting colour of the IoT device based on the detected emotional tone. After submitting the journal entry, the participant could immediately see the lighting changing. Once the journaling activity was completed, the participant filled in the second section of the questionnaire, evaluating their experience with the adaptive lighting condition.

After both scenarios were completed, the participant filled in a System Usability Scale [39] questionnaire to evaluate the overall usability of MoodUp. Right after that, a few demographic questions concluded the evaluation session.

Environmental measurements and mood visualisation features were available to the participant through the mobile app during the entire study, but they were not considered relevant outcome variables during the evaluation.

The same experimental procedure was followed for all participants to ensure consistency throughout the study.

4.4 MEASUREMENT INSTRUMENTS

The evaluation of MoodUp was focused on questionnaire responses collected after each testing condition, as mentioned in the previous subchapter. The questions were designed to point out how participants perceived the environment around them during the journaling activity and how they experienced the system’s adaptive lighting.

The main measurement instrument used throughout the study was a five-point Likert scale, where 1 represented ‘Strongly Disagree’ and 5 represented ‘Strongly Agree’. This type of scale is usually used in user experience and HCI research because it allows participants to express different levels of agreement with a statement in a simple and consistent manner. Figure 4.2 shows an example of a questionnaire item evaluated using the Likert scale. The rest of the items can be found in the questionnaire form [40].

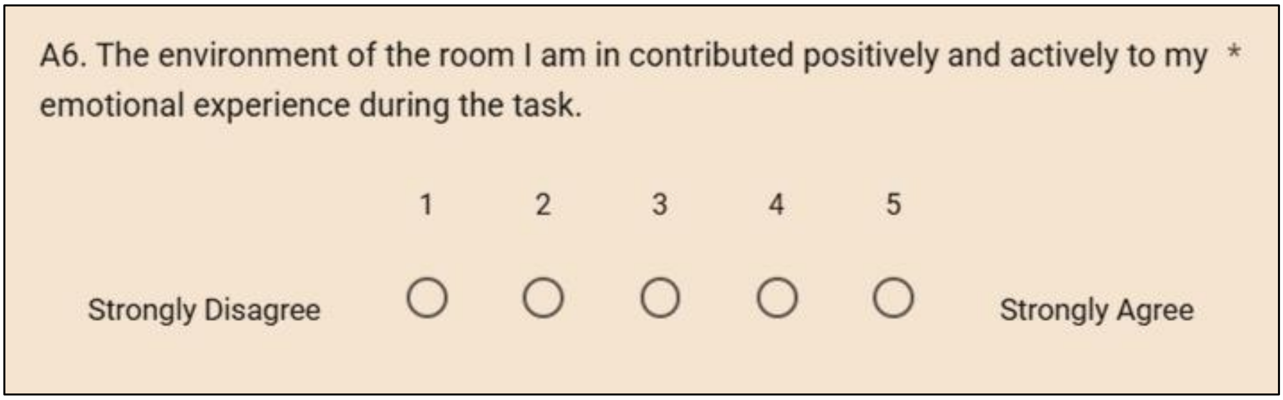

Figure 4.2 - Example of a Questionnaire Statement Using the Likert Scale

Following both testing conditions, participants answered a series of questions related to their experience. These questions focused on environmental responsiveness, emotional comfort, support during self-reflection and the overall influence of the lighting environment on the journaling activity. The same set of statements was used both in Scenario A and in Scenario B to show a direct comparison between the traditional room lighting condition and the adaptive MoodUp lighting condition.

The participants also filled in the SUS [39] questionnaire at the end of the experiment to evaluate the usability of the developed system. This scale is a standardised ten-item survey used to determine how participants perceived usability of a specific system like MoodUp. This way, the proposed system could be compared against already established usability thresholds.

4.5 DATA COLLECTION AND ANALYSIS

The data used in this evaluation was collected with the help of Google Forms. After all participants completed the study questionnaire, the responses were exported to Google Sheets to be processed.

The analysis followed a quantitative approach. For both scenarios (Scenario A and Scenario B), the Likert-scale responses were converted into numerical values from 1 to 5 (1 represented 'Strongly Disagree' and 5 represented 'Strongly Agree'). Then, the mean values were calculated for each questionnaire item under both testing conditions to highlight the differences between the traditional lighting environment and the adaptive MoodUp environment. The focus was on aspects related to emotional reflection, perceived emotional support, environmental responsiveness and the overall journaling experience.

Responses from the SUS questionnaire were analysed based on its standard scoring procedure. As half of the questions were worded positively and the other half negatively, the statements alternated throughout the questionnaire. For the positive ones, 1 point was subtracted from the participant's response, and for the negative ones, the value of the response was subtracted from 5. Then, the scores were added up in a sum, and the result was multiplied by 2.5, resulting in a final SUS score (from 0 to 100).

In addition to the numerical analysis, participants had the opportunity to provide optional comments at the end of the questionnaire, alongside the demographic questions. These answers were used to support the interpretation of the results presented in the following chapter.

4.6 ETHICAL CONSIDERATIONS

Participation in the MoodUp evaluation study was entirely voluntary. All participants were informed about the purpose of the study before taking part and they were free to stop the evaluation at any point if they no longer wished to continue.

As the study involved writing short journal entries describing personal experiences, participants were instructed to share only information they felt comfortable with. Even though the journaling activities focused on emotional experiences, the study was not intended to diagnose, treat, or evaluate any mental health conditions. The proposed system was evaluated as a tool designed to support emotional reflection and user interaction, with wellbeing improvement in mind.

There was no personally identifiable information collected during the study beyond basic demographic data. All responses were stored securely and used strictly for academic research and analysed in an anonymised form, ensuring individual identities remained completely private in the reported results.

5. RESULTS

5.1 PARTICIPANTS DEMOGRAPHIC AND BASELINE PROFILE

A total of N = 17 participants took part in the evaluation study of MoodUp.

Before jumping right into analysing the results obtained from the testing scenarios, it is useful to first look at an overview of the participant group and the demographic characteristics of the collected dataset.

Figure 5.1 shows the age distribution of the participants. As presented, the majority of participants (64.7%) belonged to the 21-25 age group. The remaining age categories had a smaller number of participants, with 11.8% for people between the ages of 31-40 and 51-60, and 5.85% for people under 20 or between 26-30.

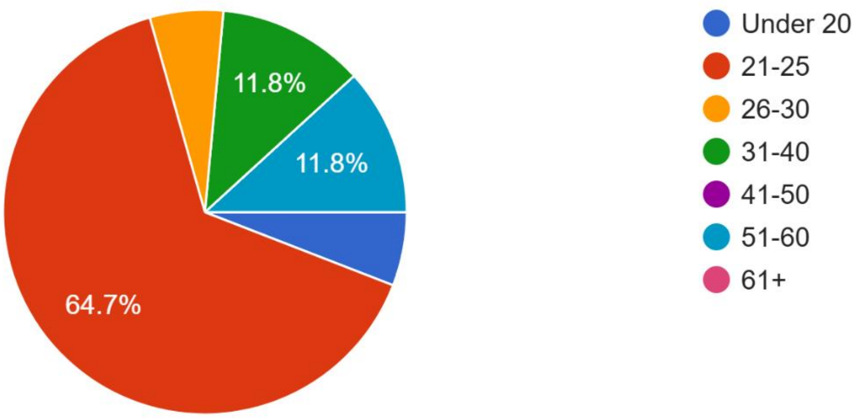

Figure 5.1 - Age Distribution of the Study Participants

The gender distribution of the participant group is shown in Figure 5.2. There were more female participants (58.8%) than male participants (41.2%) in the sample.

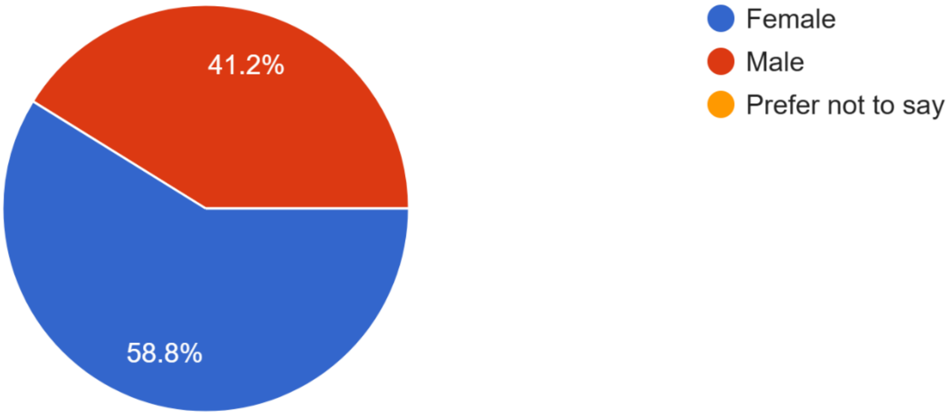

Figure 5.2 - Gender Distribution of Study Participants

Table 5.1 summarises the main demographic characteristics of the group and provides context for the interpretation of the results presented in the following section.

Table 5.1 - Summary of Participant Characteristics

Characteristic	Result
Total participants	17
Previous experience with smart lights	8 (47.1%)
No previous experience with smart lights	9 (52.9%)
Interested in emotional wellbeing technologies	13 (76.5%)
Maybe interested in emotional wellbeing technologies	4 (23.5%)

5.2 RESULTS FROM SCENARIO A: EXISTING ROOM LIGHTING

Scenario A evaluated participant perception while using the MoodUp mobile application under existing room lighting conditions. The developed adaptive lighting device remained inactive during this phase. The environment did not respond to the content of the journal entries.

Figure 5.3 presents the average scores obtained for each item during Scenario A.

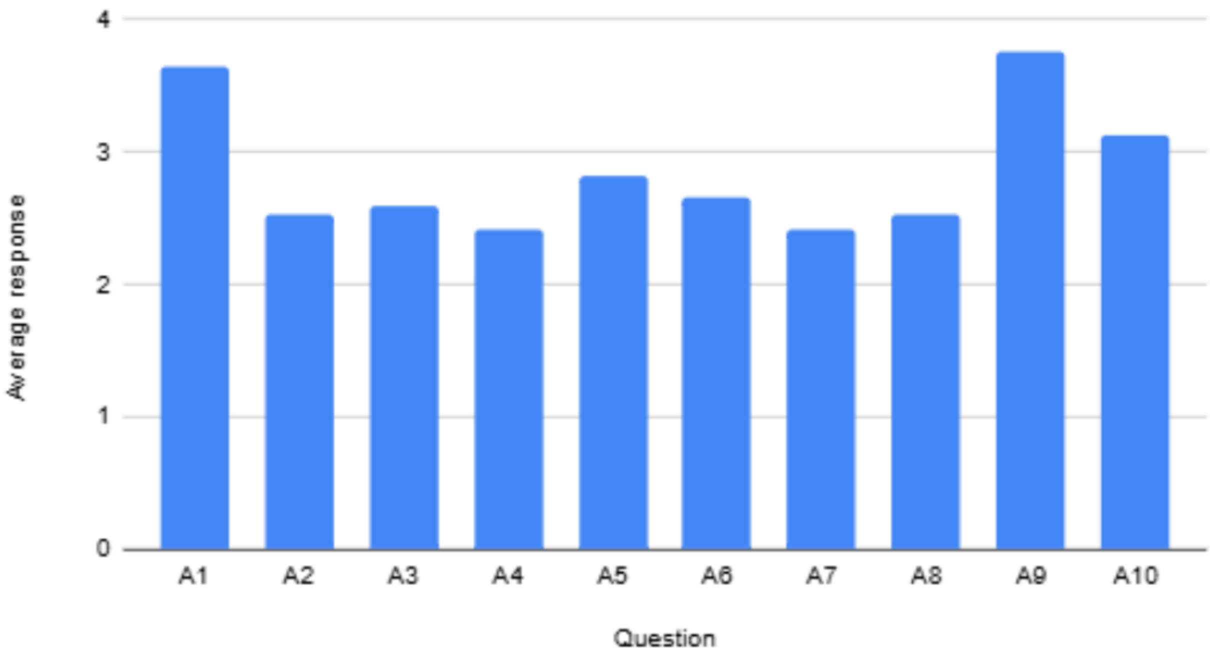

Figure 5.3 – Average Questionnaire Scores Obtained During Scenario A

The average results ranged from 2.41 to 3.76 on the five-point Likert scale. The highest score was recorded for question A9 (M = 3.76), followed by question A1 (M = 3.65). In contrast, questions A4 and A7 obtained the lowest average scores (M = 2.41).

Questions related to the journaling activity and the completion of the task obtained some of the highest scores during this scenario. Question A10 also received a score above the midpoint of the scale (M = 3.12). Lower scores were recorded for questions related to environmental responsiveness, emotional support and lighting adaptation, including A2, A7 and A8.

These results represent the participant responses obtained under existing room lighting conditions and serve as the reference values for comparison with Scenario B.

5.3 RESULTS FROM SCENARIO B: ADAPTIVE EMOTION-AWARE LIGHTING

Scenario B evaluated participant perception while using the MoodUp mobile application and the adaptive lighting component active. The system analysed the sentiment expressed in the submitted journal entries during this phase, then it automatically adjusted the lighting colour according to the detected emotional tone.

Figure 5.4 presents the average scores obtained for each item during Scenario B.

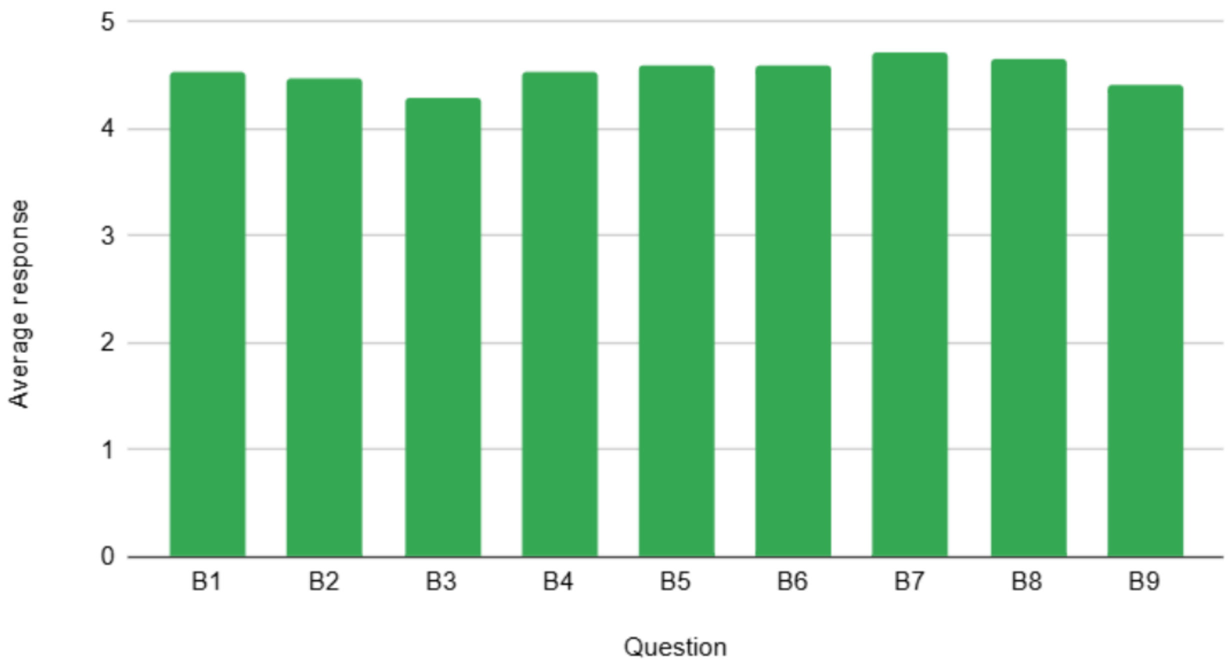

Figure 5.4 - Average Questionnaire Scores Obtained During Scenario B

The average scores ranged from 4.29 to 4.71 on the five-point Likert scale. The highest score was recorded for question B7 (M = 4.71), followed by question B8 (M = 4.65). Questions B5 and B6 both obtained an average score of 4.59. The lowest score was recorded for question B3 (M = 4.29).

Questions associated with environmental responsiveness, emotional support and lighting adaptation obtained some of the highest scores during this scenario. In contrast, question B3 recorded the lowest average score, although it remained above 4.0 on the five-point scale.

All questionnaire items obtained average scores above 4.0. These results represent the participant responses obtained under the adaptive MoodUp lighting condition and are compared with the Scenario A results in the following section.

5.4 COMPARISON BETWEEN SCENARIO A AND SCENARIO B

Table 5.2 presents the mean questionnaire scores obtained during both testing conditions together with the calculated differences between them. Positive differences were recorded for all questionnaire items.

Table 5.2 – Comparison of Mean Questionnaire Scores Between the Two Scenarios

Question	1	2	3	4	5	6	7	8	9	10
Scenario A	3.65	2.53	2.59	2.41	2.82	2.65	2.41	2.53	3.76	3.12
Scenario B	4.53	4.47	4.29	4.53	4.59	4.59	4.71	4.65	4.41	4.53
Difference	0.88	1.94	1.7	2.12	1.77	1.94	2.3	2.12	0.65	1.41

The largest increase was observed for Question 7 (+2.30), followed by Questions 4 and 8 (+2.12 each). Questions 2 and 6 both recorded an increase of +1.94. In contrast, the smallest increase was recorded for Question 9 (+0.65), followed by Question 1 (+0.88).

Figure 5.5 presents a visual comparison of the mean participant responses obtained under both lighting conditions.

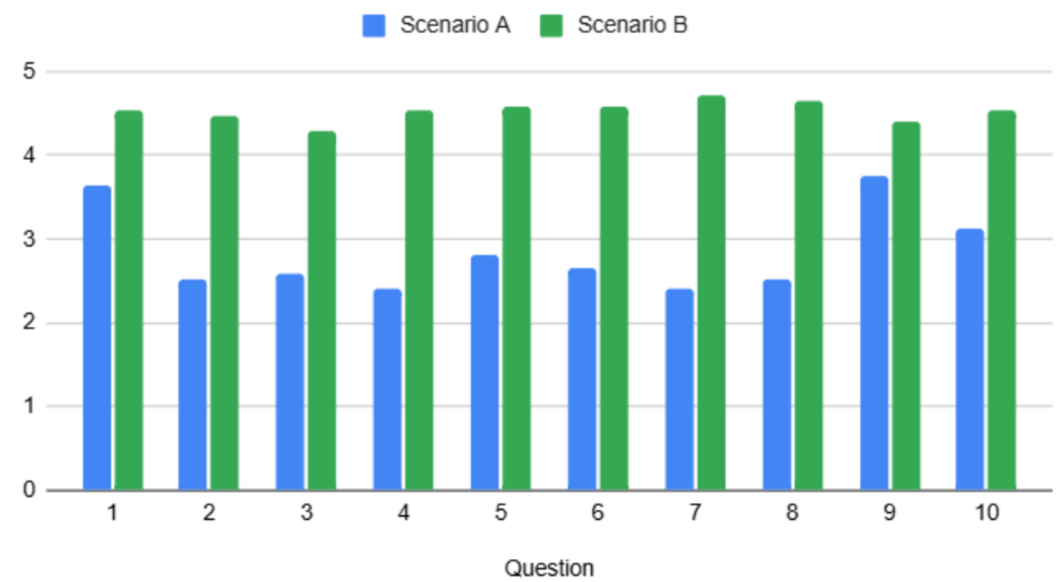

Figure 5.5 – Visual Representation of Mean Participant Scores, Scenario A vs Scenario B

For all questionnaire items, the scores recorded during Scenario B were higher than those recorded during Scenario A.

Scenario A scores ranged from 2.41 to 3.76, while Scenario B scores ranged from 4.29 to 4.71. The largest differences between the two conditions can be observed for Questions 4, 7 and 8, while the smallest differences can be observed for Questions 1 and 9.

5.5 SYSTEM USABILITY SCALE EVALUATION

At the end of the evaluation, participants completed the System Usability Scale questionnaire to assess the overall usability of MoodUp. The questionnaire consisted of ten standardised items designed to evaluate how easy a system is to learn, understand and use.

The scores were calculated according to the standard SUS procedure described in Subchapter 4.5. The individual SUS scores ranged from 40 to 100. The final average SUS score obtained by MoodUp was 90.74 out of 100, as summarised in Table 5.3.

Table 5.3 – System Usability Scale Result

Metric	**Value**
Average SUS Score	90.74
Minimum SUS Score	40
Maximum SUS Score	100

5.6 PARTICIPANTS FEEDBACK

In addition to the questionnaire responses, participants were asked which component of MoodUp would provide the greatest value if they were to use the system over a longer period of time.

Figure 5.6 presents the distribution of responses.

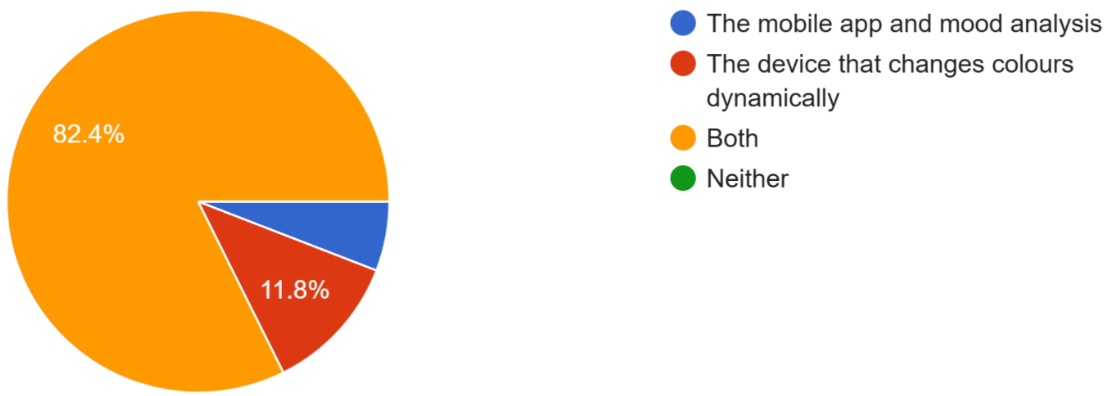

Figure 5.6 – Participant Responses Regarding the Most Valuable Component of MoodUp

The majority of participants (82.4%) selected both the mobile application and the adaptive lighting device. A smaller number of participants selected only the adaptive lighting device (11.8%) or only the mobile application (5.9%).

Participants also had the opportunity to provide optional comments regarding their experience with the system. Some comments described MoodUp as intuitive, easy to use and enjoyable to interact with. Other responses referred to the combination of digital journaling, sentiment analysis and adaptive ambient lighting. Some examples of participant feedback included: “very cool, 10/10”, “great experience”, “the best smart light I’ve ever seen”, “good work”.

6. DISCUSSIONS

6.1 OVERVIEW OF FINDINGS

The evaluation of MoodUp showed clear differences between the two lighting conditions. Participants came to the conclusion that the adaptive lighting environment more positively than the existing room lighting across all questionnaire items. The biggest differences were observed for questions related to the responsiveness of the environment, emotional adaptation and perceived support during the journaling activity.

The usability evaluation ended in positive results as well. MoodUp achieved an average System Usability Scale score of 90.74 out of 100, indicating a high level of perceived usability. Also, most participants considered the combination of the mobile application and the adaptive lighting device to be the most valuable aspect of the system.

These results indicate that mixing up digital journaling with adaptive ambient lighting creates a more responsive and emotionally supportive experience than traditional static lighting. The next sections discuss these findings in greater detail and compare them with previous research.

6.2 INTERPRETATION OF SCENARIO A

Scenario A represents the baseline condition. Participants completed the journaling activity under the existing room lighting without any adaptive environmental response. Even in these conditions, participants generally engaged positively with the activity. Question A1 received one of the highest average scores at 3.65. This suggests that writing about personal experiences encouraged emotional reflection, regardless of the lighting conditions.

The highest score in this scenario was recorded for Question A9 at 3.76. This indicates that participants found the existing lighting fitting for using the application and writing their journal entries. In other words, the room itself was comfortable enough for completing the task.

In contrast, questions related to environmental responsiveness and emotional adaptation received the lowest scores. Questions A2, A4, A7 and A8 all averaged around 2.5. This is not surprising since the lighting never changed in response to what participants wrote. As a result, the surrounding environment felt mainly like a passive setting rather than an active part of the experience.

In the end, Scenario A shows that journaling alone can encourage emotional reflection, even without adaptive environmental feedback. At the same time, it establishes a useful baseline for determining whether introducing adaptive lighting changes the way participants perceive the overall experience.

6.3 INTERPRETATION OF SCENARIO B

Unlike Scenario A, the adaptive MoodUp environment received high ratings on all questionnaire items. Every question had an average score above 4.0, showing that participants had a positive response to the adaptive lighting throughout the evaluation.

The highest scores were for Questions B7 (4.71) and B8 (4.65), which focused on how the environment responded. Participants clearly noticed that the lighting reacted to the emotions in their journal entries and generally found this adaptation meaningful. Instead of being a passive background, the lighting became part of the overall journaling experience.

Questions B5 and B6, which looked at emotional comfort and the support the environment provided during journaling, both had an average score of 4.59. This suggests that participants felt the adaptive lighting helped them while reflecting on their thoughts and feelings. One possible reason for this is that the immediate visual response strengthened the link between what participants wrote and how the environment reacted.

Another interesting point is that even the lowest score in Scenario B was above 4.0. This shows that positive evaluations were consistent across all parts of the system, not just a few specific features. Participants could use the application comfortably while also noticing and appreciating how the lighting adapted.

So, Scenario B demonstrates that adding adaptive ambient lighting changed how participants experienced the journaling activity. Compared to the static lighting condition, the environment was seen as more than just a place for the activity; it became an active part of the overall experience.

6.4 COMPARISON WITH PREVIOUS RESEARCH

The findings obtained in this dissertation are generally consistent with previous research investigating digital journaling, emotion-aware environments and usability evaluation methodologies. Table 6.1 summarises the principal findings reported in related studies (discussed in Chapter 2.5.1) and compares them with the results obtained during the evaluation of MoodUp.

Table 6.1 – Comparison of Findings Between Previous Studies and MoodUp

Study	N	Reported Findings	MoodUp Findings (current study) **N**=17
MacIsaac et al.	152	-Improved psychological wellbeing -Supported self-reflection -Positive effects of app-based journaling	-Positive engagement with journaling (A1 = 3.65) -Emotional reflection during journaling -Adaptive environment perceived as supportive
Yamauchi et al.	40	-Reduced state anxiety -Psychological reassurance -Benefits of personalised adaptation	-Positive evaluation of adaptive environment -Emotional support (B5 = 4.59) -Environmental contribution during journaling (B6 = 4.59)
Lorincz et al.	10	-SUS = 73–78 -Satisfactory perceived usability -Reliable usability assessment	-SUS = 90.74 -High perceived usability -Positive interaction with mobile and IoT components

The longitudinal study conducted by MacIsaac et al. [4] demonstrated that app-based journaling interventions can improve psychological wellbeing and support self-reflection. Similar observations were obtained in the present study. Even under existing room lighting conditions, participants engaged positively with the journaling activity and reported one of the highest scores for Question A1 (M = 3.65). Furthermore, the adaptive lighting condition was perceived as emotionally supportive, suggesting that the combination of digital journaling and environmental feedback may further reinforce reflective experiences.

Yamauchi et al. [31] reported reductions in state anxiety following emotion-aware automation and highlighted the potential of personalised adaptive environments for promoting psychological wellbeing. Although the present study did not directly measure anxiety, participants evaluated the adaptive MoodUp environment considerably more positively than the static lighting condition and reported high scores for emotional support and environmental contribution during journaling activities (B5 = 4.59; B6 = 4.59). These findings similarly suggest that environments capable of responding to emotional information may positively influence user experience and perceptions of emotional support.

Finally, Lorincz et al. [32] demonstrated the usefulness of the System Usability Scale as a reliable instrument for evaluating digital systems and reported SUS scores between 73 and 78 for their evaluated platforms. By comparison, MoodUp achieved an average SUS score of 90.74 out of 100, indicating a high level of perceived usability. Although the evaluated systems differ substantially in purpose and implementation, the findings further support the suitability of SUS for assessing emerging interactive technologies. The high score obtained by MoodUp may reflect the relatively focused scope of the application and the seamless integration between the mobile application and the adaptive lighting device, both of which were perceived positively by participants throughout the evaluation.

Table 6.2 presents a direct comparison between the System Usability Scale scores reported by Lorincz et al. [32] and the score obtained during the present evaluation.

Table 6.2 – Comparison of System Usability Scale Results

Study	Evaluated System	Participants (N)	Average SUS Score
Lorincz et al.	Figma prototype	10	78
Lorincz et al.	Live web system	10	73
MoodUp (current study)	Mobile application and adaptive lighting system	17	90.74

Taken together, previous studies have generally investigated digital journaling, emotion-aware adaptation and usability evaluation as separate research directions. The present dissertation contributes by integrating these concepts within a single platform that combines digital journaling, local sentiment analysis, environmental monitoring and adaptive ambient lighting, while also providing empirical evidence regarding user perceptions of such an emotion-aware environment.

6.5 CONTRIBUTION OF MOODUP

The main contribution of MoodUp lies in the integration of multiple technologies into a single system designed to support emotional self-reflection and user wellbeing. The proposed solution combines a mobile journaling application, sentiment analysis, environmental monitoring and an adaptive ambient lighting device that responds to emotions expressed through digital journal entries.

From a technical perspective, MoodUp demonstrates the feasibility of connecting user-centred applications with Internet of Things technologies to create environments capable of reacting to emotional information in real time. The developed prototype shows that emotional data obtained through journaling activities can be used to generate adaptive environmental responses without requiring direct user intervention.

From a practical perspective, the conducted user study provides initial evidence regarding how people perceive this type of adaptive environment. The results suggest that participants generally responded positively to the proposed system and appreciated the combination of digital journaling and adaptive ambient lighting. The findings obtained from this study may therefore serve as a starting point for future research investigating emotion-aware environments and technologies designed to support emotional reflection and wellbeing.

6.6 LIMITATIONS

Several limitations should be considered when interpreting the findings of this study. First, the evaluation involved a relatively small participant sample consisting of 17 individuals. Furthermore, participants were recruited using a convenience sampling approach and most of them belonged to a similar age range. As a result, the findings cannot be generalised to the wider population.

Another limitation is related to the duration of the evaluation. Each participant interacted with MoodUp during a single testing session lasting approximately 15–20 minutes. While this approach was sufficient for evaluating the initial user experience, it does not provide information regarding how perceptions of the system may change during long-term use.

Finally, the adaptive behaviour of MoodUp was limited to colour changes generated through sentiment analysis of journal entries. Human emotions are considerably more complex than the categories used by the proposed system and can be influenced by numerous personal and contextual factors. Moreover, the lexicon-based sentiment analysis approach employed by MoodUp does not account for contextual information, sarcasm or more nuanced linguistic patterns. Thus, the adaptive responses generated by the system should be interpreted as simplified representations of emotional states rather than precise measurements of human emotions.

7. CONCLUSIONS AND FUTURE WORK

7.1 CONCLUSION

This dissertation investigated whether combining digital journaling with adaptive ambient lighting could create a more engaging and emotionally supportive environment than traditional static lighting. To explore this, MoodUp was designed and evaluated as an emotion-aware platform that integrates digital journaling, local sentiment analysis, adaptive lighting and environmental monitoring within a single IoT system.

The user study showed clear differences between the two evaluation scenarios. Participants consistently rated the adaptive lighting condition more positively than the existing room lighting across all questionnaire items. The largest improvements were observed for environmental responsiveness, emotional adaptation and perceived support during journaling. In addition, MoodUp achieved a System Usability Scale score of 90.74 out of 100, while 82.4% of participants identified the combination of the mobile application and the adaptive lighting device as the most valuable aspect of the system.

These results suggest that adaptive environmental feedback can positively influence how users experience digital journaling. Rather than acting only as the setting in which the activity takes place, the surrounding environment became an active part of the interaction by responding to the emotions expressed in the journal entries.

Overall, the objectives of this dissertation were successfully achieved through the design, implementation and evaluation of MoodUp. Beyond presenting a fully functional prototype, this work provides initial evidence that users perceive emotion-aware adaptive environments positively and offers a foundation for future research in emotion-aware smart environments, Internet of Things technologies and human-computer interaction.

7.2 ADDRESSING THE RESEARCH QUESTIONS

The findings of this dissertation indicate that the proposed research questions were successfully addressed.

RQ1: How can a sentiment-aware IoT-based ambient lighting system be designed and implemented using digital journal entries as a source of emotional information?

This research question was addressed through the development of MoodUp, an integrated platform consisting of a Kotlin-based mobile application, Firebase cloud services and an ESP32-based IoT device. The system successfully combines digital journaling, local sentiment analysis and adaptive ambient lighting capable of responding to emotions expressed through journal entries.

RQ2: How do users perceive an adaptive ambient lighting environment that responds to emotions expressed through digital journaling?

The evaluation results demonstrated that participants generally perceived the adaptive environment positively. Questions related to environmental responsiveness, emotional adaptation and perceived support during journaling activities received some of the highest scores during the adaptive lighting condition, suggesting that participants recognised and appreciated the emotion-aware behaviour of the system.

RQ3: Is an adaptive ambient lighting environment perceived more positively than a traditional static lighting environment?

The results indicate that the adaptive lighting condition was perceived more positively than the existing room lighting condition. All questionnaire items obtained higher average scores during Scenario B, with the largest improvements observed for questions associated with environmental responsiveness and emotional adaptation.

RQ4: To what extent do users perceive MoodUp as a useful and usable tool for emotional self-reflection and wellbeing support?

The findings suggest that participants perceived MoodUp as both useful and highly usable. The system achieved an average System Usability Scale score of 90.74 out of 100, while most participants identified the combination of the mobile application and adaptive lighting device as the most valuable aspect of the system. Furthermore, participant feedback indicated that the proposed platform was considered intuitive, engaging and supportive of emotional reflection activities.

7.3 FUTURE WORK

The results obtained in this dissertation indicate that MoodUp has the potential to be developed beyond a research prototype. Participants responded positively to the adaptive environment, and the evaluation suggests that combining digital journaling with adaptive ambient lighting represents a promising direction for future emotion-aware technologies. Furthermore, current commercial solutions generally focus either on digital journaling or smart lighting, rather than integrating both within a single platform.

From a software perspective, future work could focus on extending the mobile application with additional personalisation features, richer mood visualisations and more flexible journaling capabilities. Supporting multiple mobile platforms would also improve accessibility and enable evaluation with a broader range of users.

The hardware platform could be further enhanced by integrating additional environmental sensors, such as air quality or noise monitoring, and by extending the adaptive behaviour beyond lighting alone. Integrating MoodUp with existing smart home ecosystems could allow the system to respond to emotional information by controlling multiple aspects of the indoor environment.

Another important direction involves improving the emotion recognition pipeline. Although the current lexicon-based sentiment analysis approach proved sufficient for this study, future versions could investigate machine learning or large language models capable of understanding context, sarcasm and more complex emotional expressions. At the same time, cloud-based or hybrid processing solutions could provide more advanced analysis while maintaining an appropriate balance between functionality, performance and user privacy.

Overall, this dissertation demonstrates the feasibility of an emotion-aware ambient lighting system that combines digital journaling with adaptive environmental feedback. The positive user evaluation provides a strong foundation for continuing the development of MoodUp and exploring the wider role of emotion-aware technologies in supporting emotional wellbeing and self-reflection.

7.4 WHERE AI WAS USED

In this day and age, as much as one might try, it is impossible to stay completely away from AI tools. Currently, even a simple search on Google returns an **AI Overview** before the user has even had the chance to ask for it. Every major technology company seems to advertise its own version of AI, making it difficult to avoid interacting with these tools in everyday life. Keeping this in mind, the author would like to clearly specify where AI was used during the development and writing of this dissertation.

First of all, the author is not a native English speaker, so, in times of need, AI-powered writing and paraphrasing tools were sometimes used to improve grammar, clarify ideas and suggest alternative sentence structures. The fight against writer’s block is very real, even when multiple synonym dictionaries are open at the same time. These tools helped organise thoughts and present ideas more clearly, while still leaving the final decision about wording and content to the author.

Secondly, the idea of the dissertation started from many of the author’s passions, but AI was used as a brainstorming tool during the planning stages of the project. Sometimes it is easier to ask a non-human assistant like **ChatGPT** something rather than bothering someone with stupid questions. Most of the times, AI served a role similar to that of a more experienced developer or reviewer who could offer immediate feedback at any time with no judgement. However, as with any advice, not every suggestion made the cut, the author still had the final say in what was included and what was not.

As an interesting note, the implementation of the author's bachelor's thesis included a conversational AI component. At the time, this decision was motivated by the growing popularity of AI technologies and a desire to explore their potential within smart home environments. During the development of the current dissertation, the author moved away from this approach. The focus shifted towards digital journaling and emotional self-reflection. The purpose of this shift is to encourage users to be vulnerable and express their own emotions. Introducing a conversational AI into this process could shift the user's attention away from their self-reflection and the adaptive environment to the generated conversation.

Finally, no AI tools were used to generate experimental results, participants responses, collected data or research conclusions. No generative AI was used to create diagrams, figures or images, as all visual materials were made by the author. All design decisions, implementation work, testing activities, data analysis and final interpretations were conducted by the author.

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STATEMENT REGARDING THE AUTHENTICITY OF THE THESIS PAPER *

I, the undersigned ING. POPA SORINA-ANDREEA ________________________________,
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CNP (personal numerical code) 6010428350023
author of the thesis paper MOODUP - A USER STUDY OF AN EMOTION-AWARE AMBIENT LIGHTING SYSTEM USING SENTIMENT ANALYSIS OF DIGITAL JOURNAL ENTRIES
Developed with the purpose of participating in the graduation examination completing the educational level of MASTER'S DEGREE organized by the Faculty of AUTOMATION AND COMPUTING within the Politehnica University of Timişoara, session JUNE of the academic year 2025-2026, coordonated by CONF. DR. ING. RĂZVAN BOGDAN, considering Article 34 of the *Regulation on the organization and conduct of bachelor/diploma and dissertation examinations*, approved by Senate Decision no. 109/14.05.2020, and knowing that in the event of subsequent finding of false statements, I will bear the administrative sanction provided by Art. 146 of Law no. 1/2011 – law of national education, namely the cancellation of the diploma of studies, I declare on my own responsibility that:

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